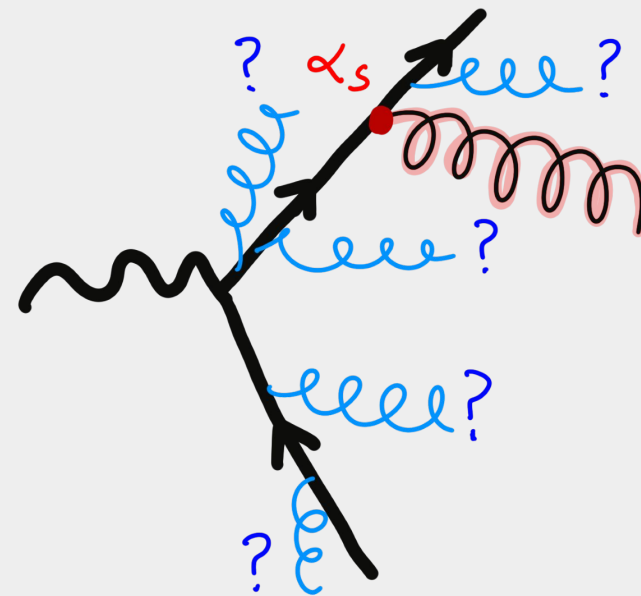


Resummed Distribution Functions: Unitary Completions of Fixed-Order Perturbation Theory



Fermilab
Theory Seminar

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In Collaboration with: Radha Mastandrea

Try our repo!

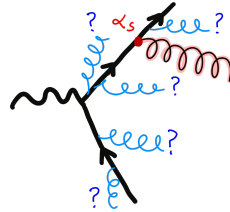
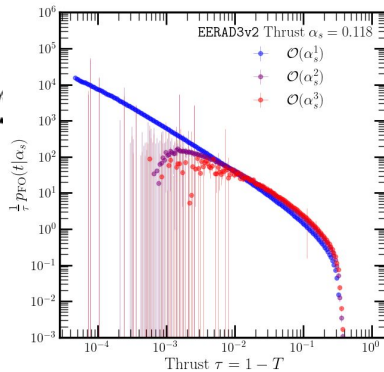


Today's Theme: *ML-inspired* QCD Theory

To me, machine learning is a *philosophy*.

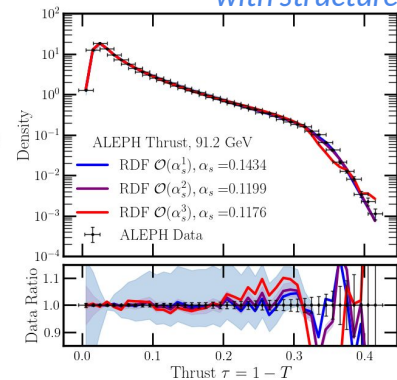
Translate physics into questions about universal function approximators and optimization.

Step (1): *Model what you know*



Step (2): *Parameterize what you don't know with structure*

$$g(t, \alpha) = -\log \left(\sum_{m=m^*}^M \frac{\alpha^m}{m!} \left| \sum_{n=0}^N g_{mn}^* \frac{t^n}{n!} \Theta_{T_m}(t - \theta_m^*) \right|_{T_m} \right) + \sum_{m=0, n=0}^{M-m^*, N} g_{\text{Analytic}, mn} \frac{t^n \alpha^m}{n! m!} \Theta_{T_m}(t - \theta_{\text{Analytic}, m})$$



$$q(t|\alpha) = f(t, \alpha) \cdot \exp \left[- \int_0^t dt' f(t', \alpha) \right]$$

Step (3): *Mesh them together*

All *good* machine learning follows these three steps.

By the end, it won't even look like machine learning anymore!

There will be ~zero neural networks or any other ML in this talk. The ML philosophy and some numerica are all we need!

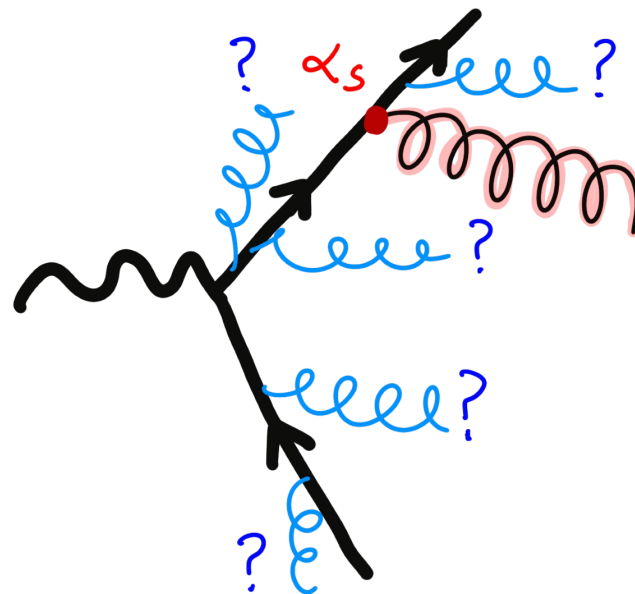
I am also interested in ML tagging for colliders, ML anomaly detection, optimal transport and geometry, and other fun applications of ML and statistics to colliders and QCD. Let's chat about it!

Basic Premise

FACT 1: Rikab is not very good at hardcore QCD calculations. Resummation is *hard*!

Large logs? IR Divergences?
Resummation? SCET? Plus
functions? Sudakov Shoulders?
Multiple scales? Too much for me!

Can I just guess?



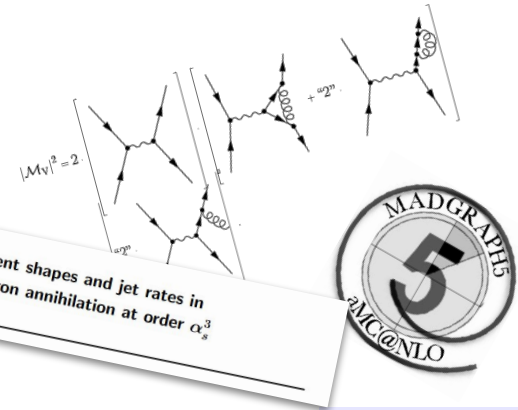
Goal: I want to see how far I can get predicting distributions for QCD observables anyways! *Including uncertainty, and as systematically and rigorously as possible!*

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Suppose I can calculate
the *fixed-order*
distribution for
observable(s) x :



EERAD3: Event shapes and jet rates in
electron-positron annihilation at order α_s^3

$$p(x|\alpha) = p_0(x) + \alpha p_1(x|\alpha) + \alpha^2 p_2(x|\alpha) + \dots + \alpha^M p_M(x|\alpha) + \mathcal{O}(\alpha^{M+1})$$

What I know: Fixed-order calculation

Unknown



FACT 2: He at least knows what a probability distribution *should* look like.

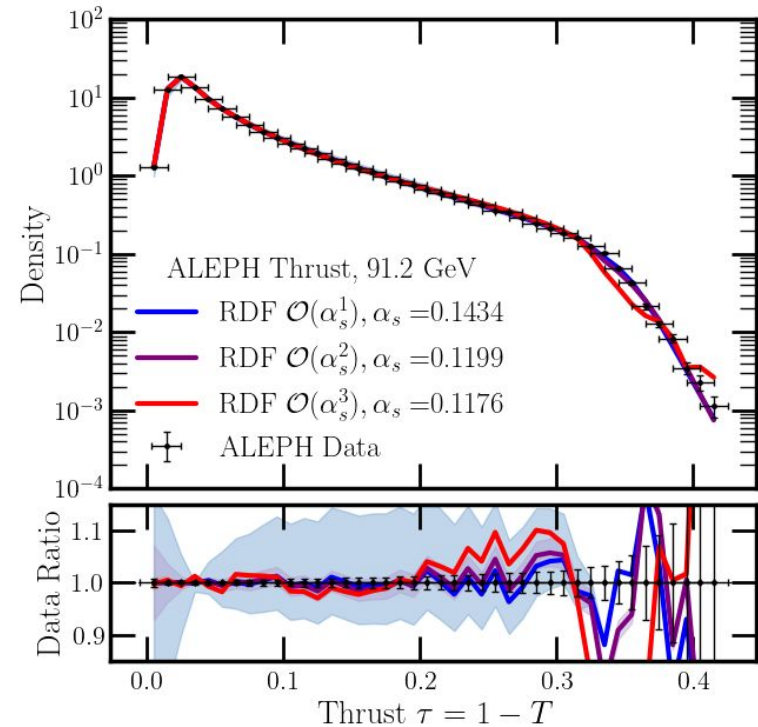
How far can we get with just *guessing* and enforcing consistency conditions?
What is the full distribution even *allowed* to be?

Let's parameterize it: The **Resummed Distribution Function**

What we are going to do

1. We will make *all-orders* statements about QCD, using only fixed-order structure and consistency conditions, *and no other knowledge!*
2. We will build a framework – the **Resummed Distribution Function (RDF)** – for encoding all possible higher-order terms, which is particularly useful for *defining theory uncertainties*.
3. We will use the RDF and above theory uncertainties to perform fits to ALEPH data – where we will get *reasonable values for α* using *only fixed order calculations* up to $\mathcal{O}(\alpha^3)$.

We don't have to know *anything* about resummation, logarithms, SCET etc, for this to work.



Order	RDF Prior σ		
	$\sigma = 0.5$	$\sigma = 1.0$	$\sigma = 2.0$
Thrust $\mathcal{O}(\alpha_s^1)$	$0.1555^{+0.0189}_{-0.0139}$	$0.1434^{+0.0409}_{-0.0212}$	$*0.19^{+0.0000}_{-0.0595}$
Thrust $\mathcal{O}(\alpha_s^2)$	$0.1202^{+0.0010}_{-0.0009}$	$0.1199^{+0.0019}_{-0.0020}$	$0.1249^{+0.0067}_{-0.0053}$
Thrust $\mathcal{O}(\alpha_s^3)$	$0.1202^{+0.0002}_{-0.0001}$	$0.1176^{+0.0002}_{-0.0002}$	$0.1164^{+0.0003}_{-0.0002}$

Basic Premise – Continued

$$p(x|\alpha) = p_0(x) + \alpha p_1(x|\alpha) + \alpha^2 p_2(x|\alpha) + \dots + \alpha^M p_M(x|\alpha) + \mathcal{O}(\alpha^{M+1})$$

Known Fixed-order calculation

Unknown



Goal: Use consistency conditions, *and nothing else*, to say *something* about this *with uncertainty!*

Consistency Conditions

The full, all-orders distribution (nature) should be:

1. Positive
2. Finite e.g. no “large logs”
3. Normalized
4. Smooth*

A fixed-order calculation is typically **none** of these things. Let's enforce it!

Basic Premise – Continued

$$p(x|\alpha) = p_0(x) + \alpha p_1(x|\alpha) + \alpha^2 p_2(x|\alpha) + \dots + \alpha^M p_M(x|\alpha) + \mathcal{O}(\alpha^{M+1})$$

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Goal: Use consistency conditions, *and nothing else*, to say *something* about this *with uncertainty!*

We introduce the **Resummed Distribution Function (RDF) q** :

$$q(t|\alpha) = f(t, \alpha) \cdot \exp \left[- \int_0^t dt' f(t', \alpha) \right]$$

Explained Shortly!

Which parameterizes everything the **unknown** terms *could ever be*, subject to the consistency conditions!

This also provides a framework for fitting and perturbative theory uncertainties.

Basic Premise – Spelled Out

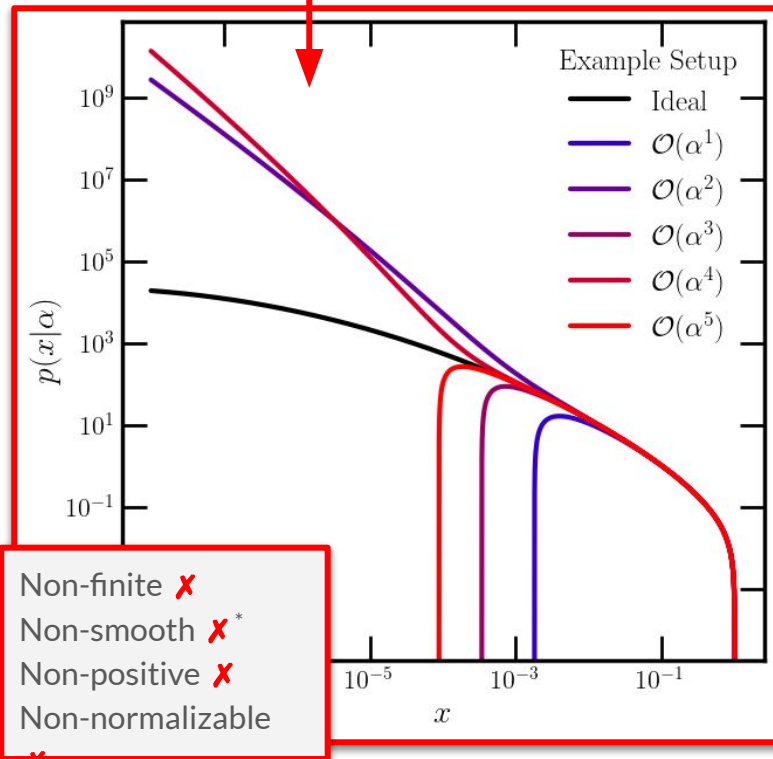
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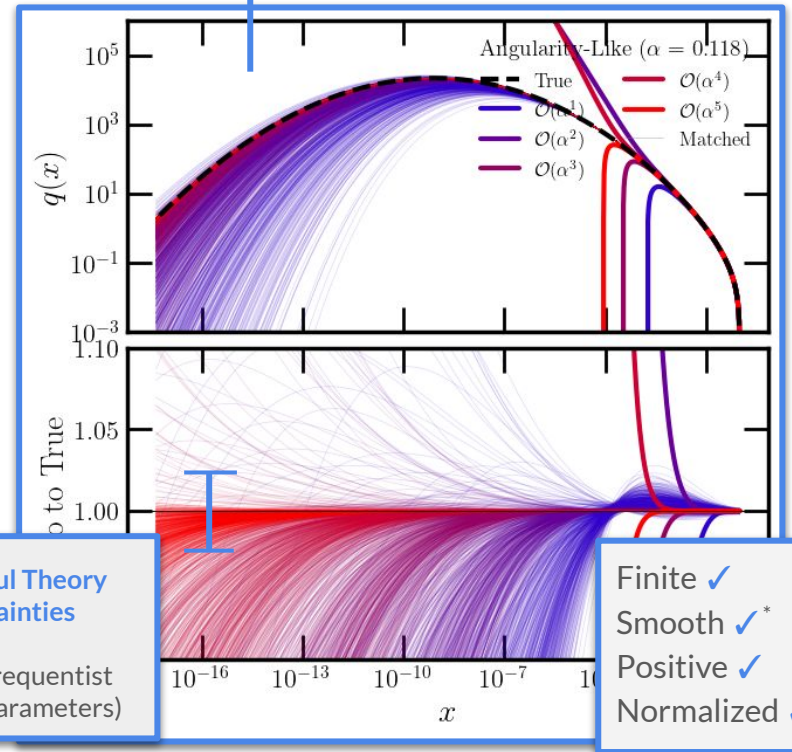
You give us this

We give you back this



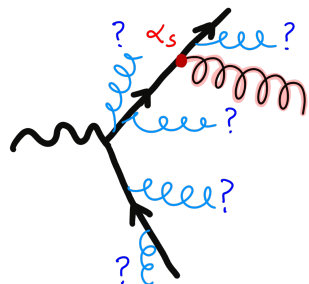
Fixed-order distribution p in α for an observable x

Matching
To RDF



All non-pathological all-orders distributions q that match p to the given order

Outline



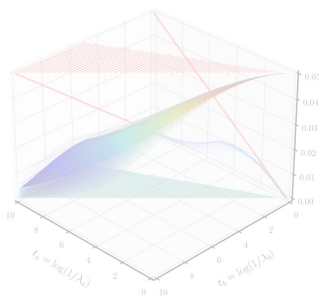
0. Basic Premise

Goal: Parameterize higher-order terms for the distribution of an observable

$$q(t|\alpha) = f(t, \alpha) \cdot \exp \left[- \int_0^t dt' f(t', \alpha) \right]$$

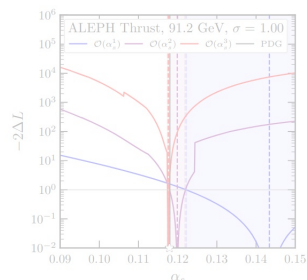
1. The Resummed Distribution Function

Given a fixed order perturbative calculation, write down a form for all possible higher order corrections consistent with unitarity



2. QCD, Examples, and Analyses

Connections to Resummation, QCD observables, numeric matching, and discussions.



3. Theory Uncertainties, Nuisance Parameters, and Fits

Fits to ALEPH data, α_s extraction, theory nuisance parameters

Constructing the **RDF**

Our goal is to parameterize the probability of measuring an observable x conditioned on a perturbative parameter α

Consistency Conditions

1. Positive
2. Finite e.g. no “large logs”
3. Normalized
4. Smooth

Perturbativity Condition

5. Analytic in α^*

Parameterize *all* such distributions!

Comes solely from **unitarity**: our observable prediction should be a proper probability!

In fact, much weaker than unitarity, which implies quantum correlation structure

Similar vibes to positivity constraints on Wilson Coefficients [[0602178](#), [1702.0613](#)]

Comes from our desire to **inject knowledge** about perturbative fixed order calculations exactly.

It is nontrivial to generically parameterize analytic condition probability distributions!

To our knowledge, has never been done before.

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Parameterize *all* such distributions!

The RDF (univariate)

Claim: Any distribution of the form

$$q(t|\alpha) = f(t, \alpha) \cdot \exp \left[- \int_0^t dt' f(t', \alpha) \right]$$

where:

- $f(t, \alpha) = \exp[-g(t, \alpha)]$
- The function g is bounded from above in t .
- The function g is analytic in α up to single logs.

$$t \equiv \log\left(\frac{1}{x}\right)$$

satisfies conditions **(1)-(5)** for any function g .

By parameterizing g , we parameterize all possible all-orders distributions!

[Details] Checking the **Conditions**

Our goal is to parameterize the probability of measuring an observable x conditioned on a perturbative parameter α

Claim: Any distribution of the form

$$q(t|\alpha) = f(t, \alpha) \cdot \exp \left[- \int_0^t dt' f(t', \alpha) \right]$$

This is just the derivative
of a simple CDF:

$$Q(t|\alpha) = 1 - e^{-\int_0^t dt' f(t', \alpha)}$$

where:

Ensures positivity!

$$f(t, \alpha) = \exp[-g(t, \alpha)]$$

$$t \equiv \log\left(\frac{1}{x}\right)$$

Ensures integration to 1!

The function g is bounded from above in t .

The function g is analytic in α up to single logs.

Ensures analyticity in α
Ensures finiteness and
smoothness depending
on the dependence of g
on t

[Aside] One can also show the converse:
All distributions can be represented for
some choice valid of g .

[Details] Checking the Conditions

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$$t \equiv \log\left(\frac{1}{x}\right)$$

The function g is bounded from above in t .
The function g is analytic in α up to single
logs.

For an observable x from 0
to 1, we transform to t
from 0 to ∞

$$t \equiv \log\left(\frac{1}{x}\right)$$

This is where our QCD logs come from!

Parametrizing the RDF

Our goal is to parameterize the probability of measuring an observable x conditioned on a perturbative parameter α

Choosing the g function

We may generically write:

$$g(t, \alpha) = -\log(g^*(t, \alpha)) + g_{\text{Analytic}}(t, \alpha)$$

positive and analytic

analytic

Each g function is just a power series in α .

We are free to choose *any*^{*} t dependence we like:
polynomials, functional bases, neural networks...

Claim: Any distribution of the form

$$q(t|\alpha) = f(t, \alpha) \cdot \exp \left[- \int_0^t dt' f(t', \alpha) \right]$$

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Our parameterization: **Polynomials**

We choose the simplest thing:

Kinematic Thresholds

$$g^*(t, \alpha), g_{\text{Analytic}}(t, \alpha) = \sum_{m=0, n=0}^{M, N} g_{mn} \alpha^m t^n \Theta(t - \theta_{mn})$$

Where each g_{mn} is a parameter matrix:

$$g_{mn} = \begin{matrix} & \xrightarrow{\text{Powers of logs } t} & \\ \begin{pmatrix} g_{00} & g_{01} & g_{02} & g_{03} & g_{04} & g_{05} & g_{06} & g_{07} & g_{08} & g_{09} \\ g_{10} & g_{11} & g_{12} & g_{13} & g_{14} & g_{15} & g_{16} & g_{17} & g_{18} & g_{19} \\ g_{20} & g_{21} & g_{22} & g_{23} & g_{24} & g_{25} & g_{26} & g_{27} & g_{28} & g_{29} \\ g_{30} & g_{31} & g_{32} & g_{33} & g_{34} & g_{35} & g_{36} & g_{37} & g_{38} & g_{39} \\ g_{40} & g_{41} & g_{42} & g_{43} & g_{44} & g_{45} & g_{46} & g_{47} & g_{48} & g_{49} \end{pmatrix} & \downarrow \text{Orders of } \alpha \end{matrix}$$

Known

Unknown

*Need $g_{0N} < 0$ to be bounded from above.

In many analytic cases, we have access to an exact $g_m(t)$ and therefore don't need it to be a polynomial in t

Matching and Perturbation Theory

Given a fixed order calculation, which *should* be a distribution:

$$p(x|\alpha) = p_0(x) + \alpha p_1(x|\alpha) + \alpha^2 p_2(x|\alpha) + \dots + \alpha^M p_M(x|\alpha) + \mathcal{O}(\alpha^{M+1})$$

What I know: Fixed-order calculation

Unknown

We want to inject our **theory knowledge** exactly into the RDF g parameters, so that the **free unknown coefficients** of the RDF will parameterize our ignorance about higher-order physics.

Matching

Powers of logs t									
g_{00}	g_{01}	g_{02}	g_{03}	g_{04}	g_{05}	g_{06}	g_{07}	g_{08}	g_{09}
g_{10}	g_{11}	g_{12}	g_{13}	g_{14}	g_{15}	g_{16}	g_{17}	g_{18}	g_{19}
g_{20}	g_{21}	g_{22}	g_{23}	g_{24}	g_{25}	g_{26}	g_{27}	g_{28}	g_{29}
g_{30}	g_{31}	g_{32}	g_{33}	g_{34}	g_{35}	g_{36}	g_{37}	g_{38}	g_{39}
g_{40}	g_{41}	g_{42}	g_{43}	g_{44}	g_{45}	g_{46}	g_{47}	g_{48}	g_{49}

$g_{mn} =$

Known

Unknown

Orders of α

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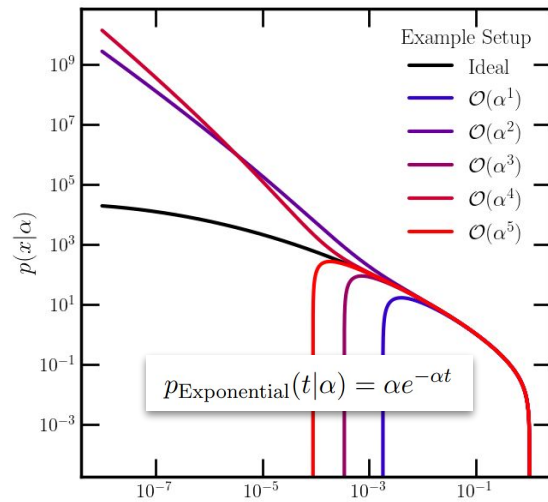
$$p(x|\alpha) = p_0(x) + \alpha p_1(x|\alpha) + \alpha^2 p_2(x|\alpha) + \dots + \alpha^M p_M(x|\alpha) + \mathcal{O}(\alpha^{M+1})$$

What I know: Fixed-order calculation

Unknown

$$p_{\text{FO}}^{(\text{Exp})}(t|\alpha) = \alpha \sum_{m=0}^{M-1} \frac{1}{m!} (-\alpha t)^m + \mathcal{O}(\alpha^{M+1})$$

Example: An α -dependent power law in x , but we only know a few orders in α



Note: Power law in x = Exponential in t

We want to inject our **theory knowledge** exactly into the RDF g parameters, so that the **free unknown coefficients** of the RDF will parameterize our ignorance about higher-order physics.

	Powers of logs t										
	Known										
$g_{mn} =$	g_{00}	g_{01}	g_{02}	g_{03}	g_{04}	g_{05}	g_{06}	g_{07}	g_{08}	g_{09}	Orders of α
	g_{10}	g_{11}	g_{12}	g_{13}	g_{14}	g_{15}	g_{16}	g_{17}	g_{18}	g_{19}	
	g_{20}	g_{21}	g_{22}	g_{23}	g_{24}	g_{25}	g_{26}	g_{27}	g_{28}	g_{29}	
	g_{30}	g_{31}	g_{32}	g_{33}	g_{34}	g_{35}	g_{36}	g_{37}	g_{38}	g_{39}	
	g_{40}	g_{41}	g_{42}	g_{43}	g_{44}	g_{45}	g_{46}	g_{47}	g_{48}	g_{49}	

Known

Unknown

Orders of α

Matching and Perturbation Theory

Given a fixed order calculation, which *should* be a distribution:

$$p(x|\alpha) = p_0(x) + \alpha p_1(x|\alpha) + \alpha^2 p_2(x|\alpha) + \dots + \alpha^M p_M(x|\alpha) + \mathcal{O}(\alpha^{M+1})$$

What I know: Fixed-order calculation

Unknown

Once we match, the resulting RDF $q(t|\alpha)$ will match the original fixed order calculation $p(t|\alpha)$ up to $O(\alpha^M)$, while also satisfying our constraints.

$$q_{\text{RDF}}(t|\alpha) = p_{\text{FO}}(t|\alpha) + \mathcal{O}(\alpha^{M+1})$$

Matching

Completely fixes the $O(\alpha^M)$ part of g !

Powers of logs t									
g_{00}	g_{01}	g_{02}	g_{03}	g_{04}	g_{05}	g_{06}	g_{07}	g_{08}	g_{09}
g_{10}	g_{11}	g_{12}	g_{13}	g_{14}	g_{15}	g_{16}	g_{17}	g_{18}	g_{19}
g_{20}	g_{21}	g_{22}	g_{23}	g_{24}	g_{25}	g_{26}	g_{27}	g_{28}	g_{29}
g_{30}	g_{31}	g_{32}	g_{33}	g_{34}	g_{35}	g_{36}	g_{37}	g_{38}	g_{39}
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$g_{mn} =$

Known

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Orders of α

Matching and Perturbation Theory [Details]

Given a fixed order calculation, which *should* be a distribution:

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What I know: Fixed-order calculation

Unknown

Analytic Matching Algorithm

$$g(t, \alpha) = -\log(p_{m^*}(t)\alpha^{m^*}) + \left[\sum_{k \geq 1} \frac{1}{k} \left(\left[-\sum_{m > m^*}^M \frac{p_m(t)}{p_{m^*}(t)} \alpha^{m-m^*} \right]^k - \left[\sum_{m=m^*}^{M-m^*} \int_0^t dt' p_m(t') \alpha^m \right]^k \right) + \mathcal{O}(\alpha^{M+1-m^*}) \right]$$

Completely fixes the $\mathcal{O}(\alpha^M)$ part of g !

Matching

Details

It is convenient to divide out the lowest power of α , denoted by m^* , so the leading term is always 1.

For exact matching, we don't need to expand g in terms of powers of t , we can just use exact functions.

By design, this calculation does not determine g_m for orders $m > M$.

Powers of logs t									
g_{00}	g_{01}	g_{02}	g_{03}	g_{04}	g_{05}	g_{06}	g_{07}	g_{08}	g_{09}
g_{10}	g_{11}	g_{12}	g_{13}	g_{14}	g_{15}	g_{16}	g_{17}	g_{18}	g_{19}
g_{20}	g_{21}	g_{22}	g_{23}	g_{24}	g_{25}	g_{26}	g_{27}	g_{28}	g_{29}
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Orders of α

Matching and Perturbation Theory [Details]

Given a fixed order calculation, which *should* be a distribution:

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What I know: Fixed-order calculation

Unknown

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Completely fixes the $\mathcal{O}(\alpha^M)$ part of g !

Matching

Powers of logs t									
g_{00}	g_{01}	g_{02}	g_{03}	g_{04}	g_{05}	g_{06}	g_{07}	g_{08}	g_{09}
g_{10}	g_{11}	g_{12}	g_{13}	g_{14}	g_{15}	g_{16}	g_{17}	g_{18}	g_{19}
g_{20}	g_{21}	g_{22}	g_{23}	g_{24}	g_{25}	g_{26}	g_{27}	g_{28}	g_{29}
g_{30}	g_{31}	g_{32}	g_{33}	g_{34}	g_{35}	g_{36}	g_{37}	g_{38}	g_{39}
g_{40}	g_{41}	g_{42}	g_{43}	g_{44}	g_{45}	g_{46}	g_{47}	g_{48}	g_{49}

$g_{mn} =$

Orders of α

Implementation

Looks scary, but autodiff magic makes this programmatically easy for a python function $p(t, \alpha)$

Try our Repo!



Matching and Perturbation Theory [Example]

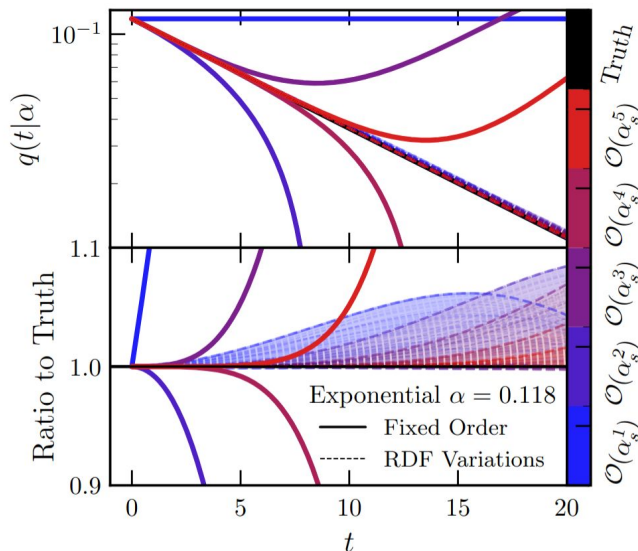
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What I know: Fixed-order calculation

Unknown

$$p_{\text{FO}}^{(\text{Exp})}(t|\alpha) = \alpha \sum_{m=0}^{M-1} \frac{1}{m!} (-\alpha t)^m + \mathcal{O}(\alpha^{M+1})$$



Applying Analytic Matching

With the RDF, we can now take a fixed order calculation and force it to be unitary by parameterizing higher order terms!

Completely fixes the $\mathcal{O}(\alpha^M)$ part of g !

Powers of logs t									
g_{00}	g_{01}	g_{02}	g_{03}	g_{04}	g_{05}	g_{06}	g_{07}	g_{08}	g_{09}
g_{10}	g_{11}	g_{12}	g_{13}	g_{14}	g_{15}	g_{16}	g_{17}	g_{18}	g_{19}
g_{20}	g_{21}	g_{22}	g_{23}	g_{24}	g_{25}	g_{26}	g_{27}	g_{28}	g_{29}
g_{30}	g_{31}	g_{32}	g_{33}	g_{34}	g_{35}	g_{36}	g_{37}	g_{38}	g_{39}
g_{40}	g_{41}	g_{42}	g_{43}	g_{44}	g_{45}	g_{46}	g_{47}	g_{48}	g_{49}

Known

Unknown Free (Nuisance) Parameters

Orders of α

We can do anything we like with the **unknown terms**: set to 0, wiggle them around randomly or [later] treat as **nuisance parameters** to capture theory uncertainty!

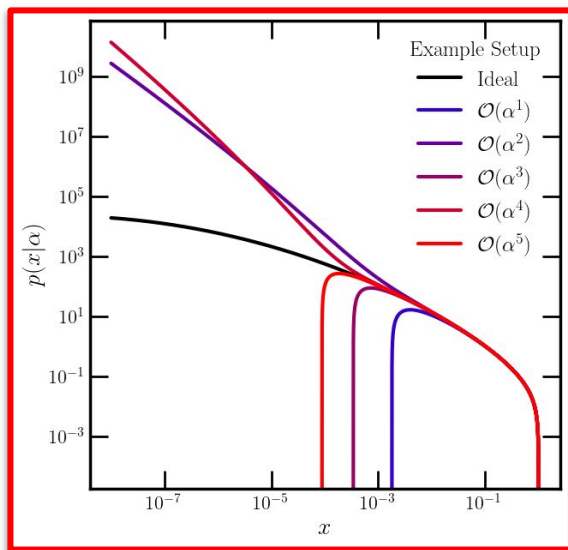
Upshot

Given a fixed order calculation, which *should* be a distribution:

$$p(x|\alpha) = p_0(x) + \alpha p_1(x|\alpha) + \alpha^2 p_2(x|\alpha) + \dots + \alpha^M p_M(x|\alpha) + \mathcal{O}(\alpha^{M+1})$$

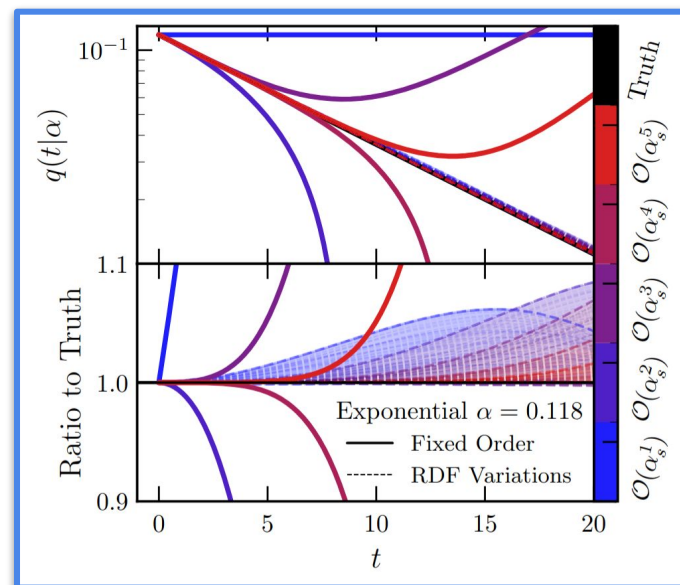
What I know: Fixed-order calculation

Unknown



The RDF lets us encode fixed-order physics we do know, and parameterize high-order physics we don't know

All while being a proper probability distribution and preserving perturbative structure exactly!



$$q^{(\text{Exp}) (M)}(t|\alpha) = g^*(t, \alpha) \exp \left[-g_{\text{Analytic}}(t, \alpha) - \int_0^t dt' g^*(t', \alpha) \exp(-g_{\text{Analytic}}(t', \alpha)) \right]$$

$$= \alpha \exp(-\alpha t) + \mathcal{O}(\alpha^{M+1}).$$

Upshot

Given a fixed order calculation, which *should* be a distribution:

$$p(x|\alpha) = p_0(x) + \alpha p_1(x|\alpha) + \alpha^2 p_2(x|\alpha) + \dots + \alpha^M p_M(x|\alpha) + \mathcal{O}(\alpha^{M+1})$$

What I know: Fixed-order calculation

Unknown

Assumptions and Disclaimers



We assume the expansion $p_m(x)$ is the fixed-order expansion of some all-orders distribution – *not always true, QFT is asymptotic!*

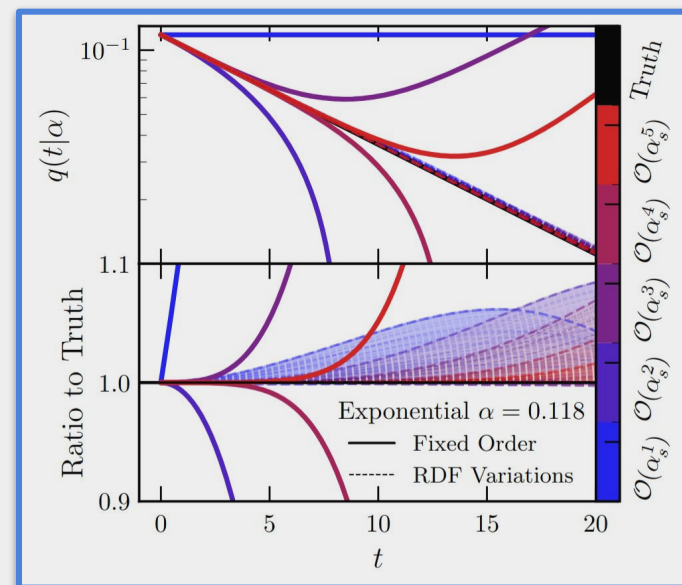
We can only capture perturbative information. No non-perturbative info!

These conditions are violated in QCD (asymptotic series, etc).

$$g_{\text{RDF}} = \sum_m \alpha^m g_m + \cancel{\frac{1}{\alpha} g_{\text{asympt.}}} + \cancel{g_{\text{non. pert.}}} ?$$

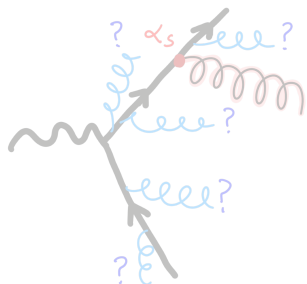
We will make the bold assumption that perturbation theory is enough.

Not Today **Not Today**
... But it could be possible!



$$x) = g^*(t, \alpha) \exp \left[-g_{\text{Analytic}}(t, \alpha) - \int_0^t dt' g^*(t', \alpha) \exp(-g_{\text{Analytic}}(t', \alpha)) \right] \\ = \alpha \exp(-\alpha t) + \mathcal{O}(\alpha^{M+1}).$$

Outline



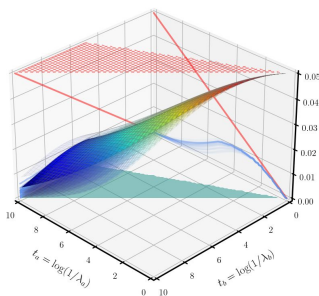
0. Basic Premise

Goal: Parameterize higher-order terms for the distribution of an observable

$$q(t|\alpha) = f(t, \alpha) \cdot \exp \left[- \int_0^t dt' f(t', \alpha) \right]$$

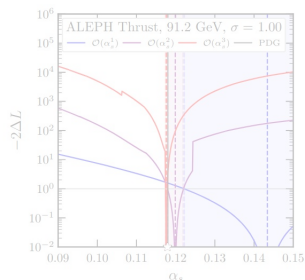
1. The Resummed Distribution Function

Given a fixed order perturbative calculation, write down a form for all possible higher order corrections consistent with unitarity



2. QCD, Examples, and Analyses

Connections to Resummation, QCD observables, numeric matching, and discussions.



3. Theory Uncertainties, Nuisance Parameters, and Fits

Fits to ALEPH data, α_s extraction, theory nuisance parameters

Resummed Distribution Functions

The RDF works by re-organizing the perturbative series from expanding in α^m to expanding in

$$\alpha^{m-m^*} \frac{p_m(t)}{p_{m^*}(t)} \ll 1$$

← Analogous, and sometimes equal to, to expanding in $\alpha^m \log^n$

Recall: α^{m^*} and p_{m^*} given by the leading order term

This is *not necessarily* the usual logarithmic NⁿLL resummation in QCD, just a **generic all-orders re-summation** consistent with unitarity.

The p_m may contain large logs, and then these will be resummed in g , but *only those particular logarithms implied by the fixed-order input* and no others.

No RG structure or factorization used whatsoever. Completely physics agnostic!

NⁿLL Resummed Distribution Functions (?)

The RDF is *an* all-orders resummation that matches the F.O. calculation. It is not *the* logarithmic resummation – but they relate!

$$P(x|\alpha) = \left(1 + \sum_{m=1}^{\infty} C_m \alpha^m\right) \times \exp \left[- \sum_{m=1}^{\infty} \sum_{n=1}^{m+1} F_{mn} \alpha^m \log^n(1/x) \right] + \cancel{R(x)}_{\text{Non-Singular}}$$

Leading Log: All terms $F_{n,n+1}$ included

Next-to-Leading Log: All terms $F_{n,n}$ included

Maps directly onto the RDF *only if* (a) f is polynomial and (b) all relevant logs are included in the F.O. calculation to match (including running coupling, NGLs, etc)

$$Q(x|\alpha) = \exp[-F(\log(1/x), \alpha)], \quad \text{where } F(t, \alpha) = \int_0^t dt' f(t', \alpha)$$

Higher order unconstrained g = “guesses” of the $N^{n>N}$ LL logarithms, can set to zero

NⁿLL Resummed Distribution Functions (?)

Takeaway

The RDF will not do NⁿLL resummation for free — that requires all-orders physics knowledge and real work!

But:

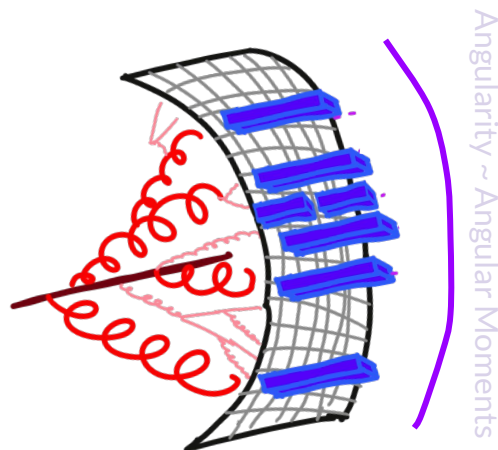
- It *does produce* an **all-orders output** consistent with unitarity as a tower of logs.
- It *does* **match to a fixed-order calculation** like an NⁿLL calculation might — one can choose to either match just the logs or the entire nonsingular part.
- It is *flexible enough* to **encode the NⁿLL calculation** for some choice of g , even if we do not know that particular g .
- It *can* **reproduce NⁿLL results** if they are either implied by the fixed order calculation or by unitarity — e.g. **Sudakov factors fixed at LL**.

All without needing to know any higher/all-order structure of QCD!

QCD-Like Toy: Pseudo-Angularity

Indulge me a game: Let's pretend the world is described by *Leading Log QCD*, but we didn't know that.

We'll predict *jet angularities* x , which (roughly) describe how wide a QCD jet is in terms of angular moments.



Annoying notation: $x = \lambda$ for angularities

$$\lambda^{(\beta)} = \sum_i z_i \left(\frac{\theta_i}{R} \right)^\beta$$

R = Jet Radius

z_i = Particle energy fraction

θ_i = Distance to center

In this world, the true (*unknown*) distribution of angularities is:

$$p_{\text{Rayleigh}}(x|\alpha) = \alpha \frac{\log\left(\frac{1}{x}\right)}{x} e^{-\alpha \frac{\log^2\left(\frac{1}{x}\right)}{2}}$$

x space: Sudakov Double Log

$$p_{\text{Rayleigh}}(t|\alpha) = \alpha t e^{-\alpha \frac{t^2}{2}}$$

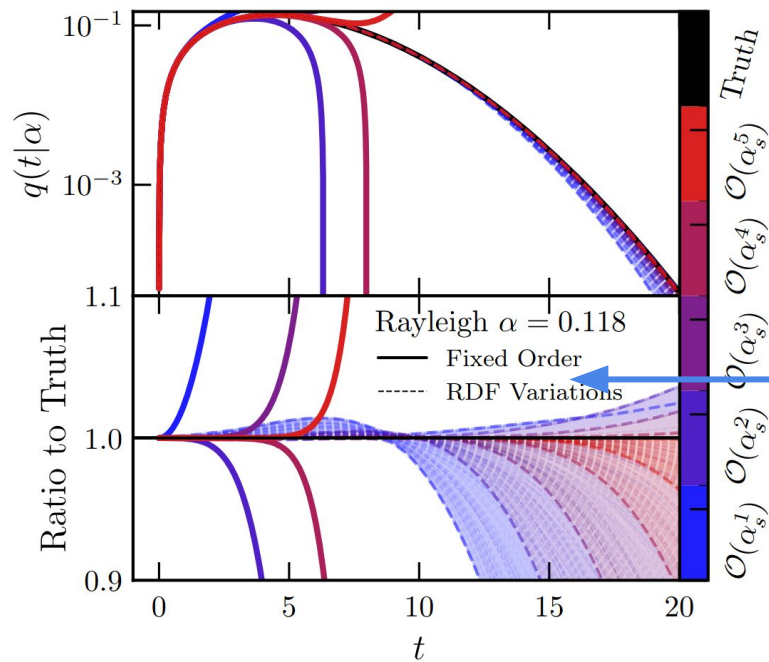
t space: Rayleigh Distribution

In this world, a fixed order QCD calculation would give us:

$$p_{\text{FO}}^{(\text{Ray})}(t|\alpha) = \alpha t \sum_{m=0}^{M-1} \frac{1}{m!} \left(-\alpha \frac{t^2}{2} \right)^m + \mathcal{O}(\alpha^{M+1})$$

QCD-Like Toy: Pseudo-Angularity

Indulge me a game: Let's pretend the world is described by *Leading Log QCD*, but we didn't know that.



The exact RDF we get is:

$$q^{(\text{Rayleigh})}(M)(t|\alpha) = \alpha t \exp\left(-\frac{\alpha t^2}{2}\right) + \mathcal{O}(\alpha^{M+1})$$

Unitarity is enough to fix the Sudakov factor at LL, exactly!

We are free to vary/guess higher-order g coefficients, corresponding to varying/guessing N^{nLL} structure.

The RDF gives well-behaved distributions from fixed order!

In this world, a fixed order QCD calculation would give us:

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As promised!

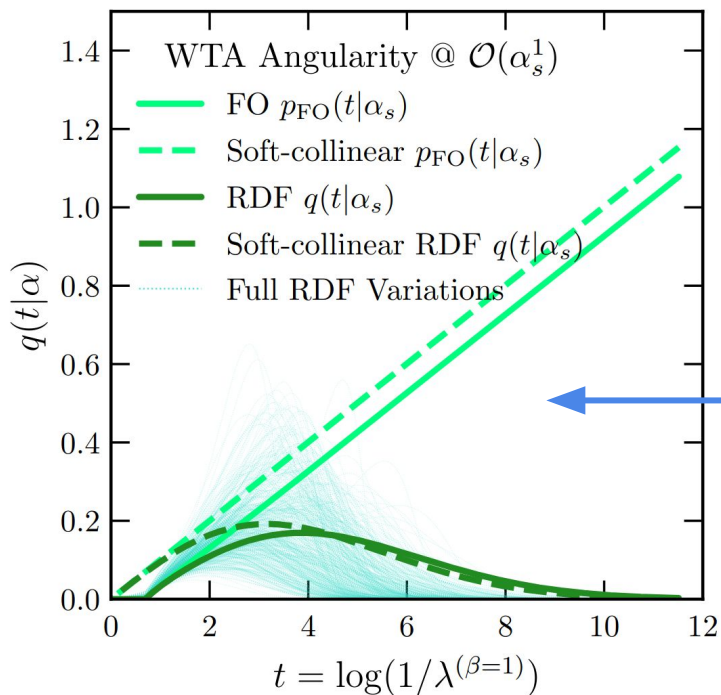
QCD Example: Real Angularities

The same game, but now with **real angularities** in real QCD, to $O(\alpha^1)$!

Technical details: “Winner-Take All” Angularities with angular exponent $\square$ in conical jets

The exact RDF we get is:

$$q(t|\alpha) = \left[\left(\frac{\alpha_s C_F}{\pi \beta} \right) \left(2t + 2 \log(1 - e^{-t}) + 3e^{-t} - \frac{3}{2} \right) \Theta(t - \log 2) \right] \\ \times \exp \left[- \left(\frac{\alpha_s C_F}{\pi \beta} \right) \left[t^2 - (\log 2)^2 + 2 (\text{Li}_2(e^{-t}) - \text{Li}_2(\frac{1}{2})) - 3(e^{-t} - \frac{1}{2}) - \frac{3}{2}(t - \log 2) \right] \right] \\ \text{(plus higher order g not shown)}$$



Applying the RDF

In our world, a fixed order QCD calculation gives us:

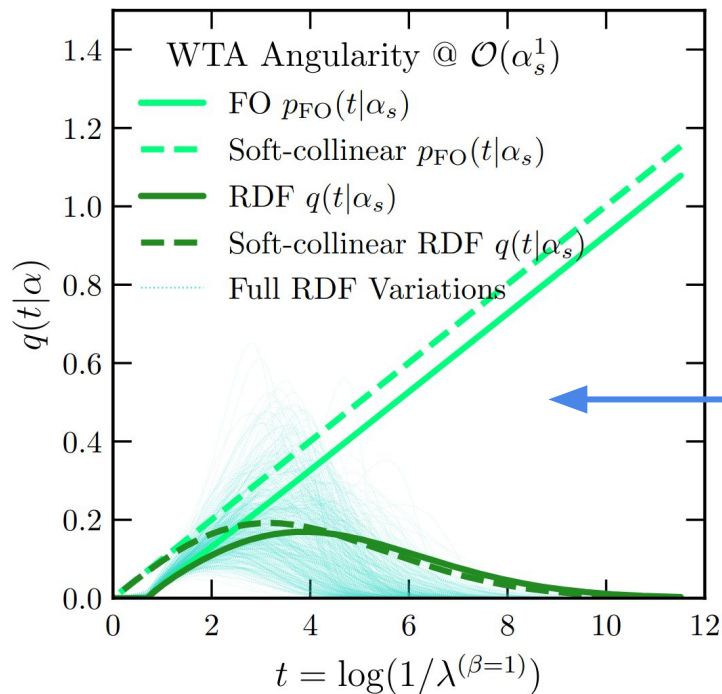
$$p_{\text{FO}}(t) = \left(\frac{\alpha_s C_F}{\pi \beta} \right) \left(2t + 2 \log(1 - e^{-t}) + 3e^{-t} - \frac{3}{2} \right) \Theta(t - \log 2)$$

As promised!

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(plus higher order g not shown because it is long)

The story barely changes! We capture the LL limit for free as t goes to zero, and can capture higher-order uncertainty.

One new complication: **Kinematic thresholds**. Angularities don't saturate $x \rightarrow 1$ at fixed order, so the RDF is cut at $t > t_0$.

Technical details: Thresholds can be encoded as a $N^1\text{LL}$ shift of the LL logs.

Applying the RDF

$$p_{\text{FO}}(t) = \left(\frac{\alpha_s C_F}{\pi \beta} \right) \left(2t + 2 \log(1 - e^{-t}) + 3e^{-t} - \frac{3}{2} \right) \Theta(t - \log 2)$$

In our world, a fixed order QCD calculation gives us:

Numeric Matching: Beyond Leading Order

Beyond leading order in QCD, fixed-order analytic calculations are unavailable.

Instead, programs like **EERAD3** or Madgraph5 produce numerical predictions.

We'll use this to generate QCD thrust to $O(\alpha^3)$

Not a problem for the RDF! Let's just alter g to make it numerically robust, then we can match it to Monte-Carlo histograms at fixed order.

We fit for coefficients g_{mn}^* and $g_{\text{Analytic},mn}$, thresholds θ_m , and temperatures T_m

$$g(t, \alpha) = -\log \left(\sum_{m=m^*}^M \frac{\alpha^m}{m!} \left| \sum_{n=0}^N g_{mn}^* \frac{t^n}{n!} \Theta_{T_m}(t - \theta_m^*) \right|_{T_m} \right) + \sum_{m=0, n=0}^{M-m^*, N} g_{\text{Analytic},mn} \frac{t^n}{n!} \frac{\alpha^m}{m!} \Theta_{T_m}(t - \theta_{\text{Analytic},m})$$

Smoothed-out absolute value to keep the log argument positive.

Factorials for numeric scaling

"Softmax" with a learned temperature to smooth thresholds

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Smoothed-out absolute value to keep the log argument positive.

$$+ \sum_{m=0, n=0}^{M-m^*, N} g_{Analytic,mn} \frac{t^n}{n!} \frac{\alpha^m}{m!} \Theta_{T_m}(t - \theta_{Analytic,m})$$

Factorials for numeric scaling

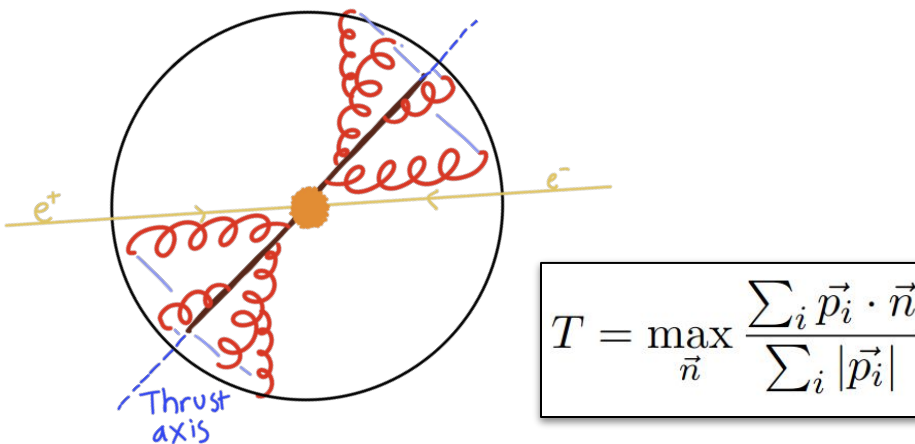
"Softmax" with a learned temperature to smooth thresholds

Fit to minimize the mean-squared error between the **Taylor expansion of q** and generated histograms for many α

Thrust and EERAD3

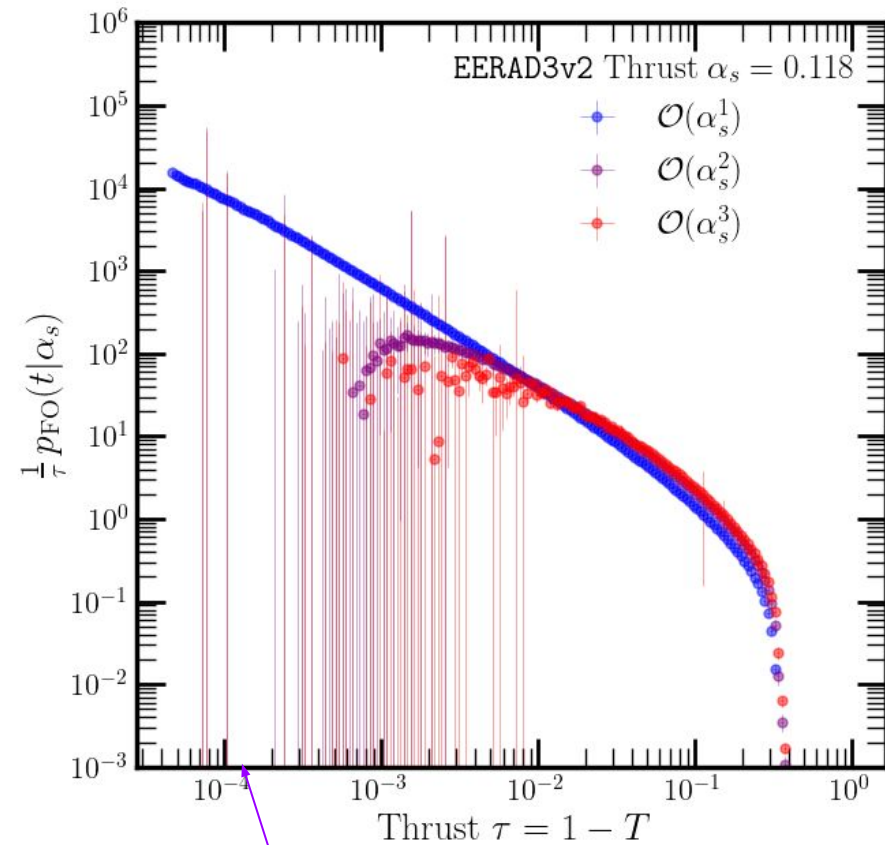
Thrust is a staple observable for precision QCD, studied to high order.

Roughly, how pencil-like is my event?



We use **EERAD3v2** to numerically calculate the fixed order distributions to $\mathcal{O}(\alpha^3)$. We get histograms of thrust for many different α values at each order.

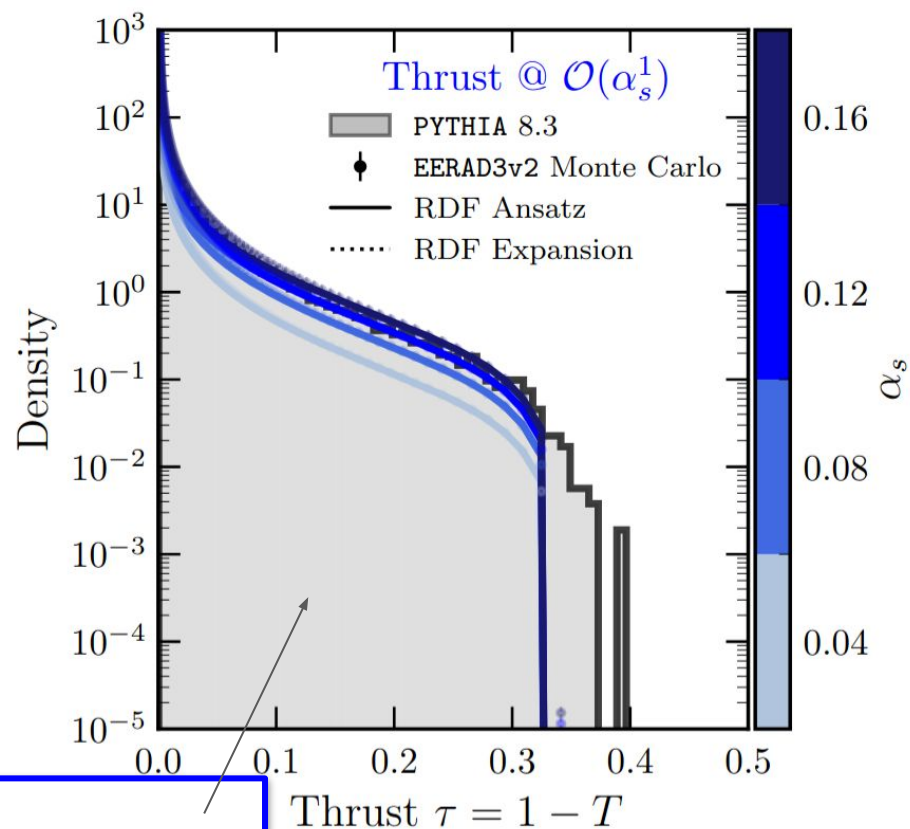
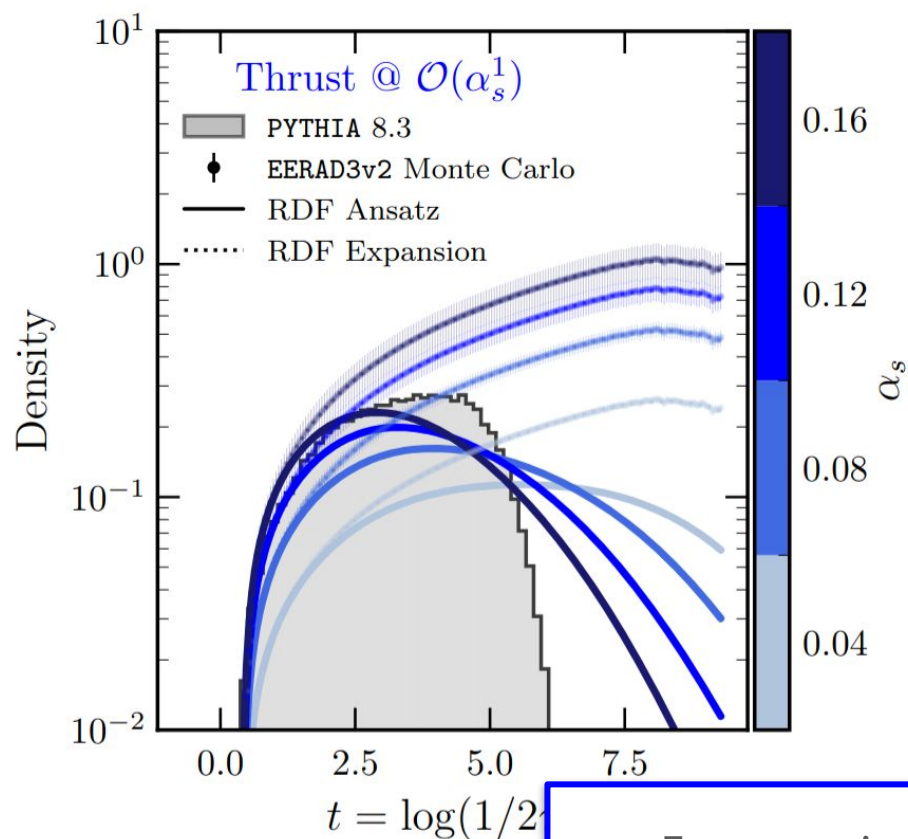
e^+e^- collisions at the Z pole



At $\mathcal{O}(\alpha^2)$, goes negative!

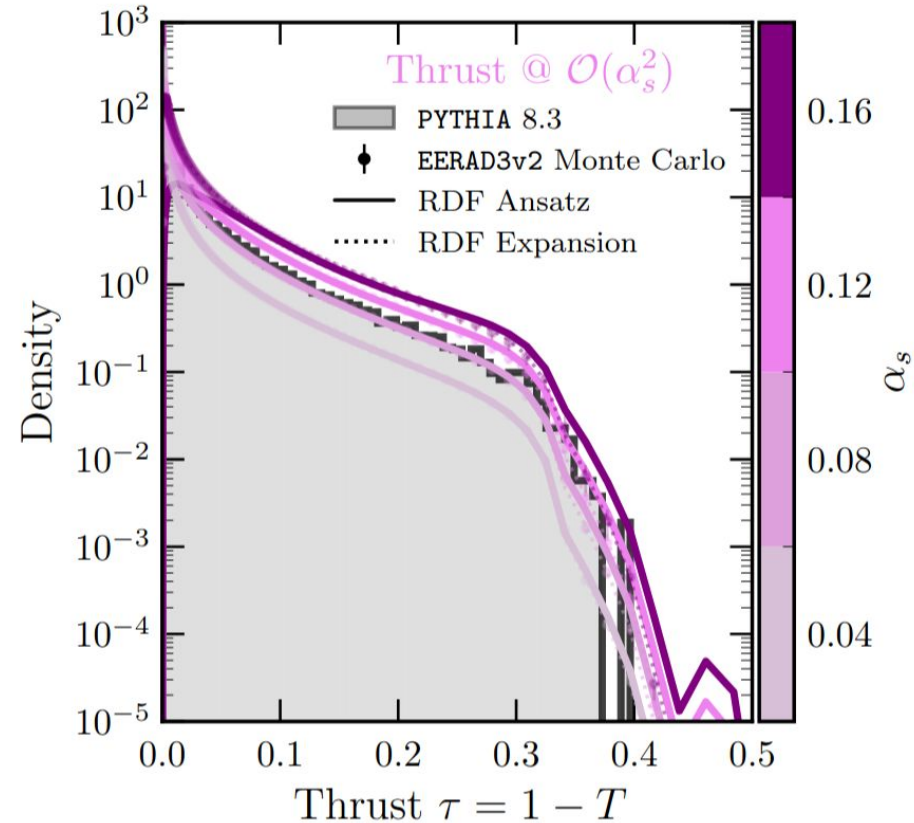
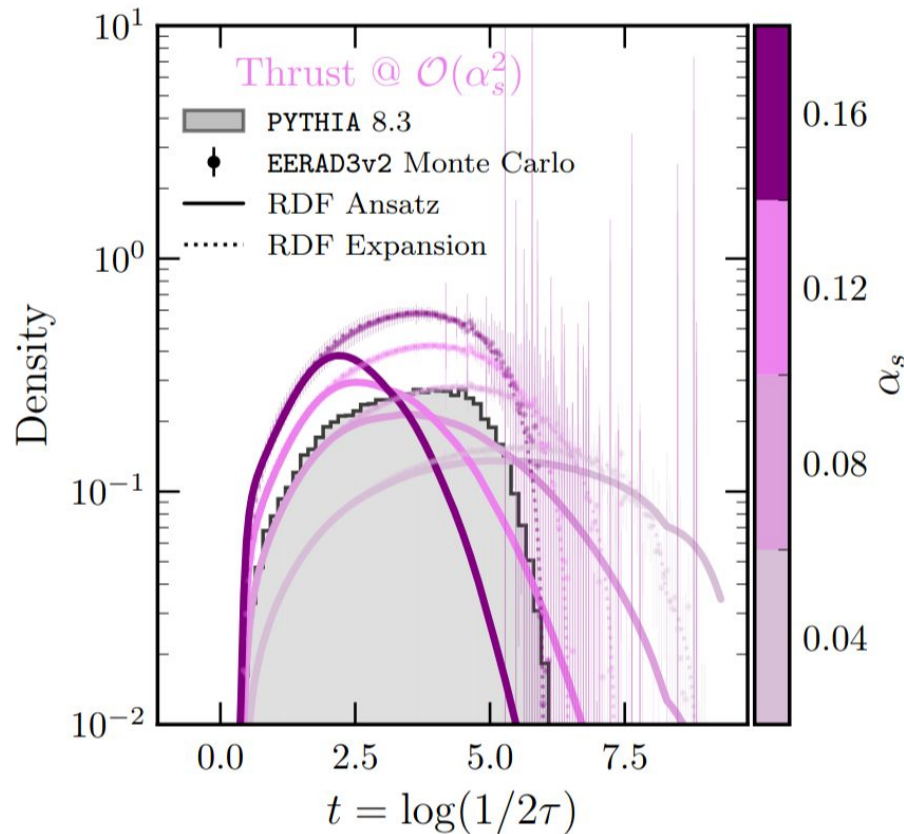
Matching Thrust to $\mathcal{O}(\alpha_s^1)$

Tune the RDF $m = 1$ parameters such that the RDF Expansion equals the Monte Carlo



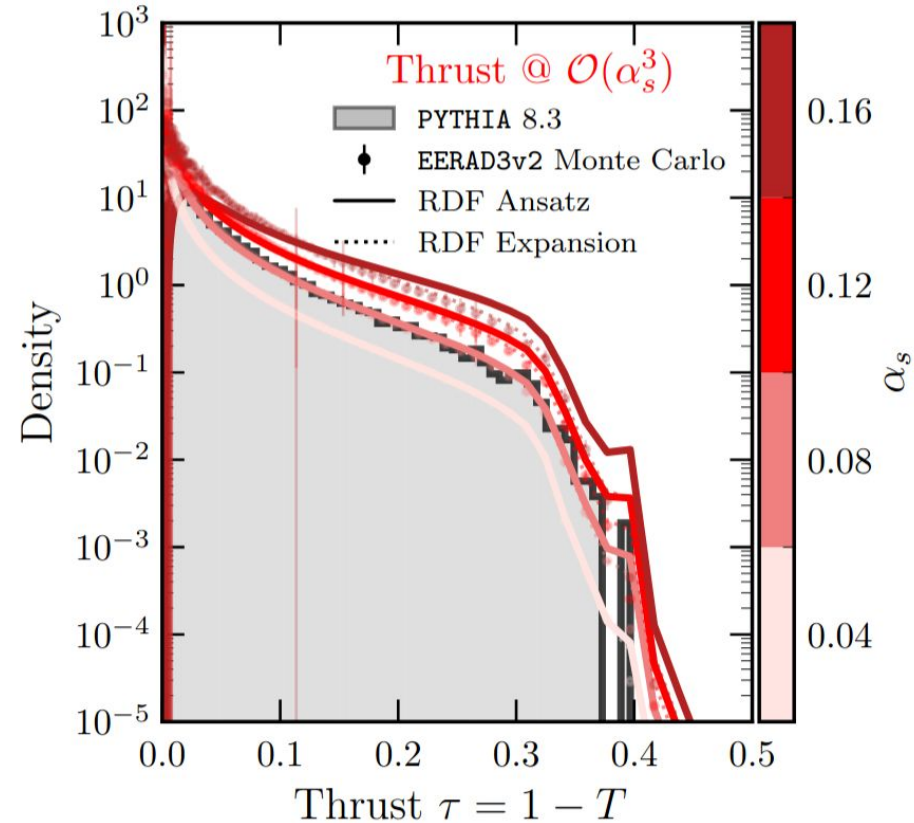
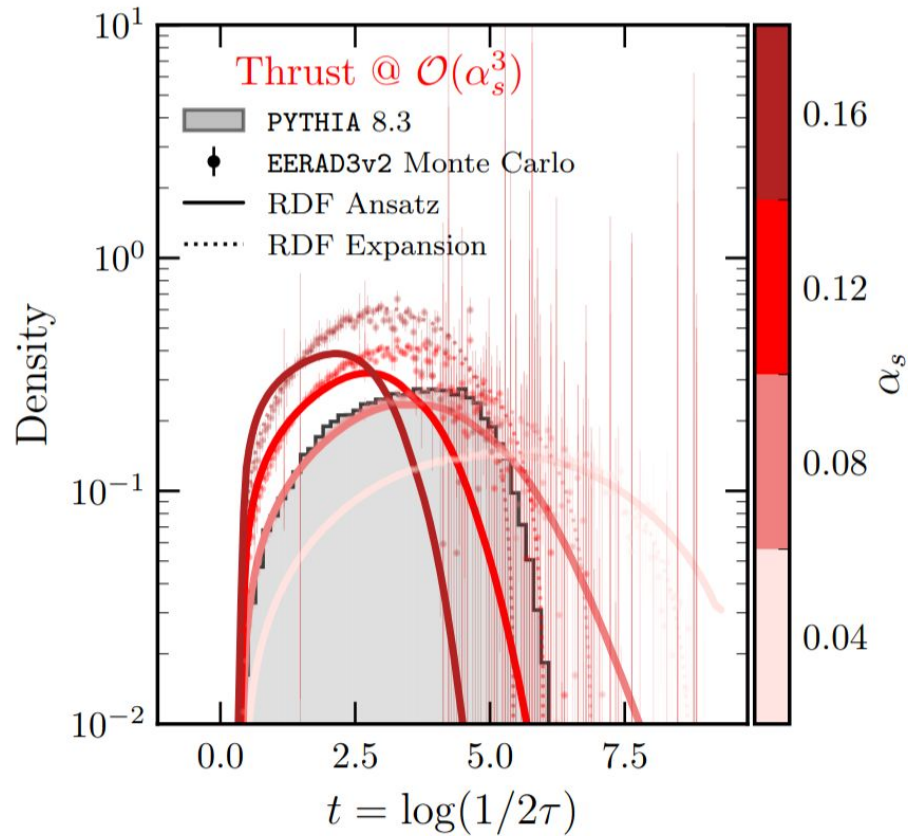
For comparison, PYTHIA
Leading Log Shower + some NLL
and nonperturbative effects

Matching Thrust to $\mathcal{O}(\alpha_s^2)$

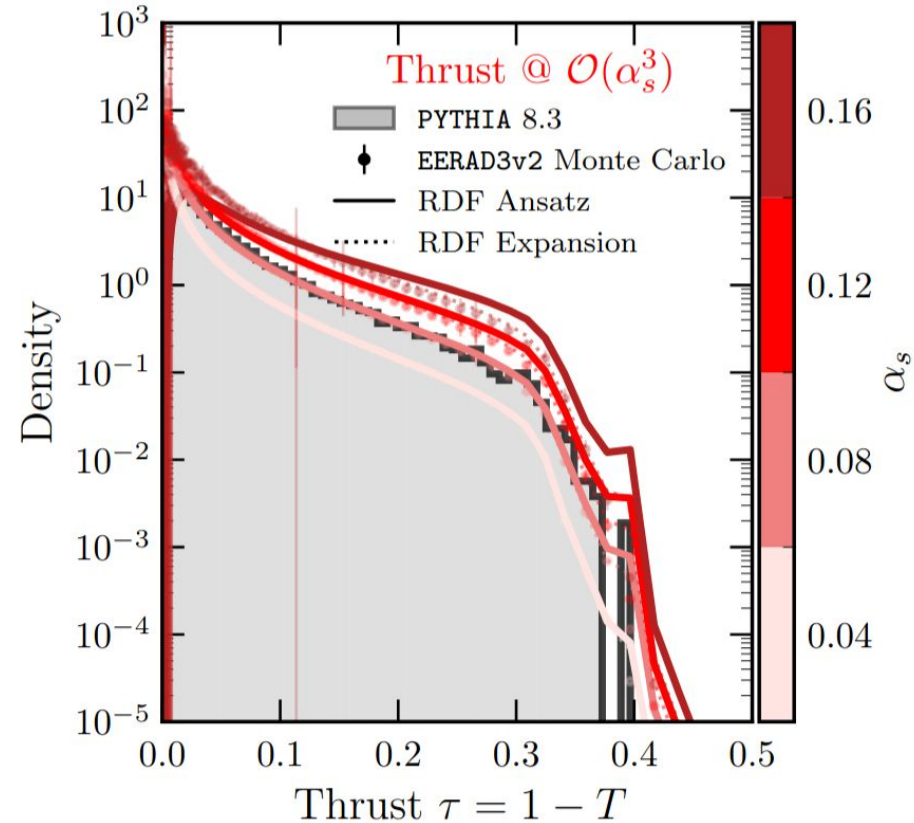
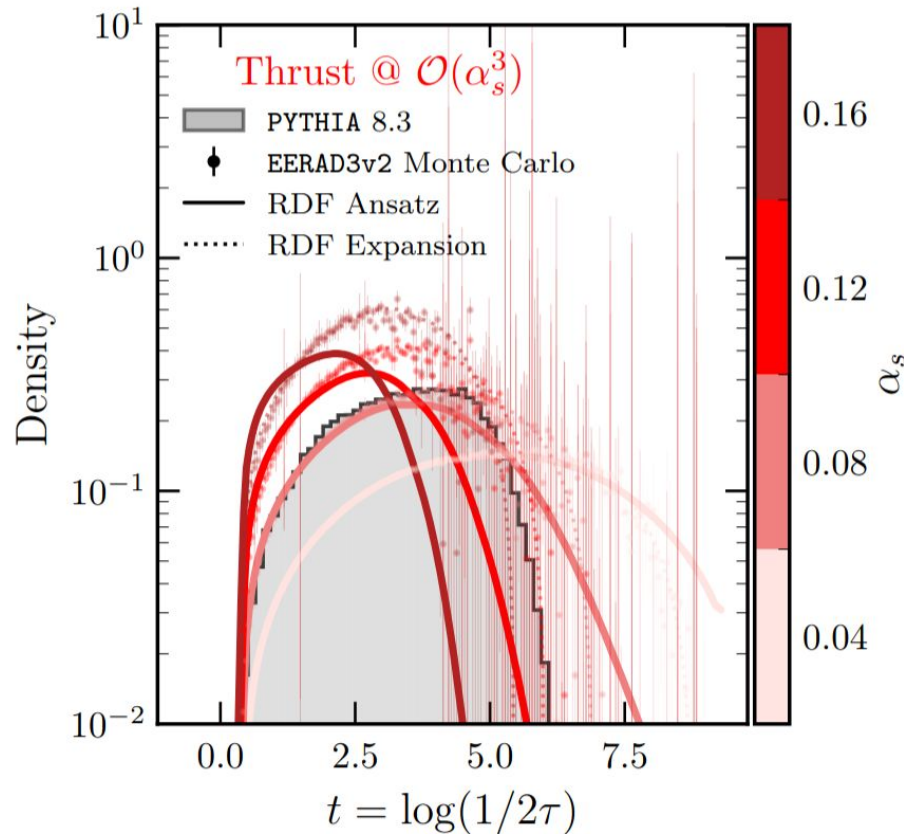


Iterative Matching: We freeze the $\mathcal{O}(\alpha_s^1)$

Matching Thrust to $\mathcal{O}(\alpha_s^3)$



Matching Thrust to $\mathcal{O}(\alpha_s^3)$



We'll keep these in our pocket for now and use this to fit to real data in a bit!

Short Detour: Simultaneous Observables

The RDF can be used to parameterize *multiple* observables at once!

$$q(t_1, t_2, \dots, t_k | \alpha) = f_1(t_1, \alpha) \times \dots \times f_k(t_1, t_2, \dots, t_k, \alpha) \\ \times \exp \left[- \int_0^{t_1} dt'_1 f_1(t'_1, \alpha) \right] \times \dots \times \exp \left[- \int_0^{t_k} dt'_k f_k(t_1, t_2, \dots, t'_k, \alpha) \right]$$

This is just a product of:

1. The RDF of observable t_1 given α
2. The RDF of observable t_2 given α and t_1
3. ...
4. The RDF of observable t_k given α and $t_{k-1}, t_{k-2}, \dots, t_1$

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Follows straightforwardly from decomposing multivariate distributions

“**autoregressively**”: $p(t_1, t_2, \dots, t_k | \alpha) = p(t_1 | \alpha) \cdot p(t_2 | t_1, \alpha) \cdot \dots \cdot p(t_k | t_1, t_2, \dots, t_{k-1}, \alpha)$

Short Detour: Simultaneous Observables

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3. ...
4. The RDF of observable t_k given α and $t_{k-1}, t_{k-2}, \dots, t_1$

This *does* depend
on the observable
ordering ... but only
at higher orders!



At the matching order, the labeling
doesn't matter. Good!

Follows straightforwardly from decomposing multivariate distributions

“**autoregressively**”: $p(t_1, t_2, \dots, t_k | \alpha) = p(t_1 | \alpha) \cdot p(t_2 | t_1, \alpha) \cdot \dots \cdot p(t_k | t_1, t_2, \dots, t_{k-1}, \alpha)$

Simultaneous Observables: Joint Angularity

Let's try angularities with two different angular moment weights!

$$\begin{aligned}
 p(t_\alpha, t_\beta) &= \frac{\alpha_s C_F}{\pi(\alpha - \beta)} \\
 p(t_\beta) &= \frac{\alpha_s C_F}{\pi\beta} t_\beta \Theta(1 - e^{-t_\beta}) \\
 p(t_\alpha|t_\beta) &= \frac{\beta}{\alpha - \beta} \frac{1}{t_\beta} \Theta(t_\beta < t_\alpha < \frac{\alpha}{\beta} t_\beta)
 \end{aligned}$$

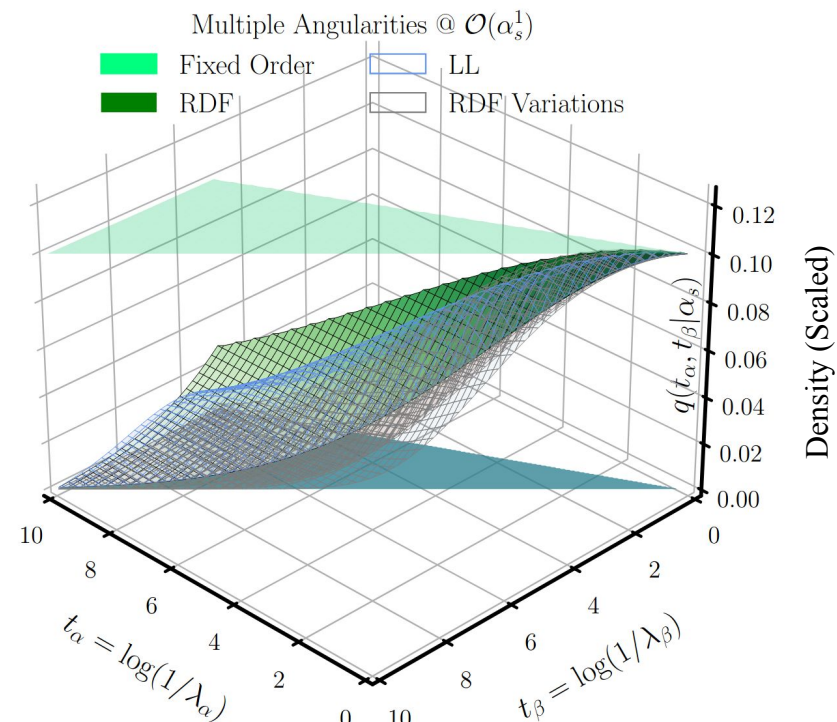
For simplicity,
soft-collinear limit
of angularities to
leading order



Turning the RDF crank...

$$q(t_\alpha, t_\beta|\alpha_s) = \frac{2\alpha_s C_F}{\pi(\alpha - \beta)} \left(1 + \underbrace{\mathcal{O}(\alpha_s)}_{\text{Higher Order unconstrained terms}} e^{-\alpha_s \frac{C_F}{\pi\beta} t_\beta^2} - \mathcal{O}(\alpha_s) \Theta_{\alpha\beta} \right)$$

Joint resummations are generically hard.
But with the RDF, we can at least begin to
parameterize what they might look like!



*Need to be extremely careful with Theta-Functions!

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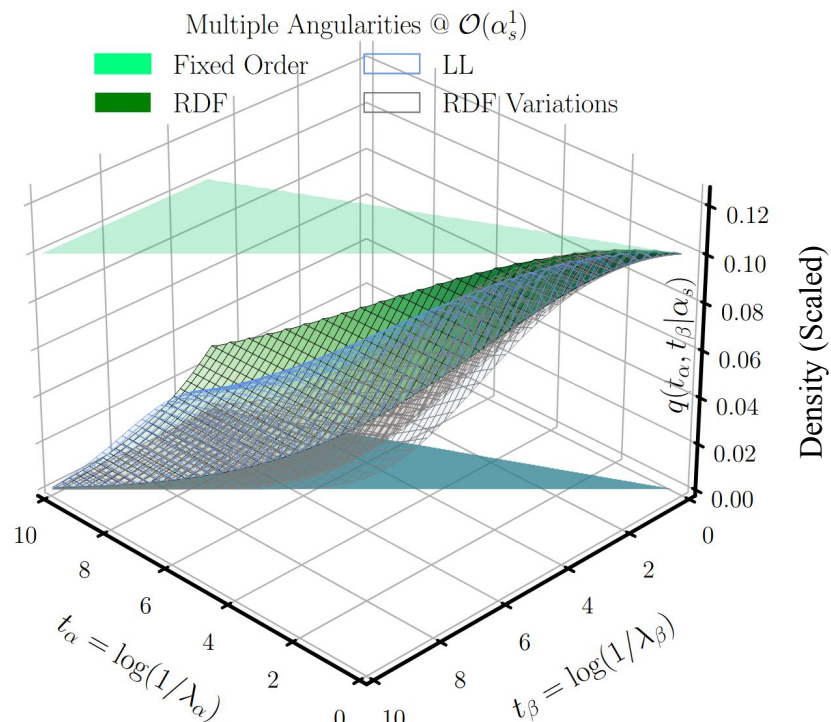
Higher Order unconstrained terms

Let's look at the marginals:

$$q(t_\beta|\alpha_s) = \frac{2\alpha_s C_F}{\pi\beta} t_\beta e^{-\frac{\alpha_s C_F}{\pi\beta} t_\beta^2}$$

$$q(t_\alpha|\alpha_s) = \frac{2\alpha_s C_F}{\pi(\alpha - \beta)} \frac{\sqrt{\pi}}{2\sqrt{\frac{\alpha_s C_F}{\pi\beta}}} \left[\operatorname{erf}\left(\sqrt{\frac{\alpha_s C_F}{\pi\beta}} t_\alpha\right) - \operatorname{erf}\left(\frac{\beta}{\alpha} \sqrt{\frac{\alpha_s C_F}{\pi\beta}} t_\alpha\right) \right]$$

Not an exact match ... but its ok to $O(\alpha^1)$! We can make the match perfect by fiddling with higher-order g to cancel out these effects



Simultaneous Observables: Joint Angularity

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For simplicity,
soft-collinear limit
of angularities to
leading order



Turning the RDF crank...

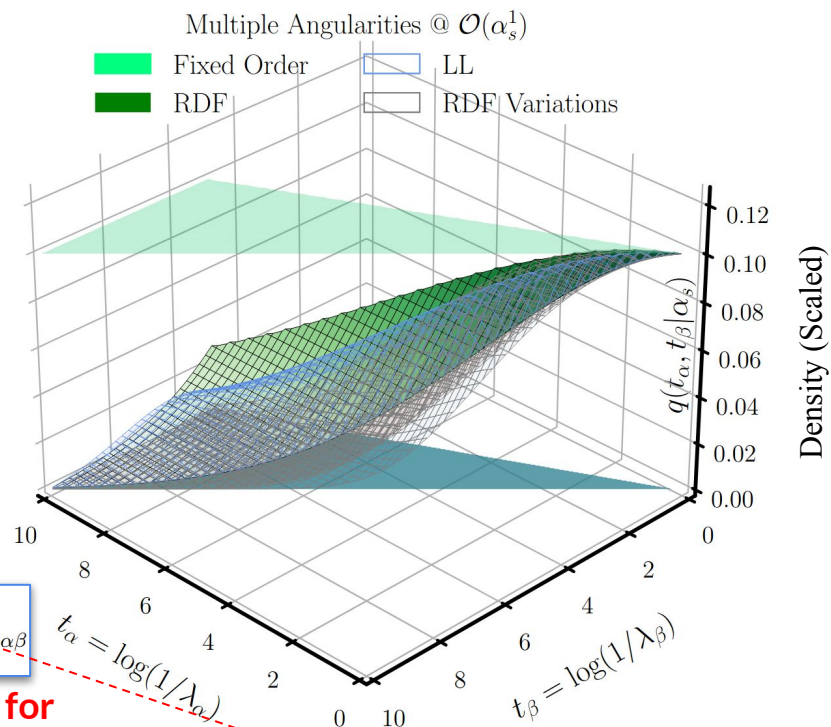
$$q(t_\alpha, t_\beta|\alpha_s) = \frac{2\alpha_s C_F}{\pi(\alpha - \beta)} (1 + \mathcal{O}(\alpha_s)) e^{-\alpha_s \frac{C_F}{\pi\beta} t_\beta^2} \mathcal{O}(\alpha_s) \Theta_{\alpha\beta}$$

Let's compare to the known LL:

$$q^{\text{LL}}(t_\alpha, t_\beta|\alpha_s) = \frac{2\alpha_s C_F}{\pi(\alpha - \beta)} \left(1 + \frac{2\alpha_s C_F}{\pi(\alpha - \beta)} \frac{(t_\beta - t_\alpha)(\beta t_\alpha - \alpha t_\beta)}{\beta} \right) e^{-\alpha_s \frac{C_F}{\pi} \left(\frac{t_\beta^2}{\beta} + \frac{(t_\alpha - t_\beta)^2}{\alpha - \beta} \right)} \Theta_{\alpha\beta}$$

**This log is
unaccounted for**

**These logs
match**



The RDF is *not* guaranteed to be a full logarithmic resummation! **This log** is only first visible at $\mathcal{O}(\alpha^2)$, so the RDF matched to $\mathcal{O}(\alpha^1)$ missed it.

Simultaneous Observables: Joint Angularity

Let's try angularities with two different angular moment weights!

$$\begin{aligned} p(t_\alpha, t_\beta) &= \frac{\alpha_s C_F}{\pi(\alpha - \beta)} \\ p(t_\beta) &= \frac{\alpha_s C_F}{\pi\beta} t_\beta \Theta(1 - e^{-t_\beta}) \\ p(t_\alpha|t_\beta) &= \frac{\beta}{\alpha - \beta} \frac{1}{t_\beta} \Theta(t_\beta < t_\alpha < \frac{\alpha}{\beta} t_\beta) \end{aligned}$$

For simplicity,
soft-collinear limit
of angularities to
leading order



Turning the RDF crank...

$$q(t_\alpha, t_\beta|\alpha_s) = \frac{2\alpha_s C_F}{\pi(\alpha - \beta)} (1 + \mathcal{O}(\alpha_s)) e^{-\alpha_s \frac{C_F}{\pi\beta} t_\beta^2} \mathcal{O}(\alpha_s) \Theta_{\alpha\beta}$$

Let's compare to the known LL:

$$q^{\text{LL}}(t_\alpha, t_\beta|\alpha_s) = \frac{2\alpha_s C_F}{\pi(\alpha - \beta)} \left(1 + \frac{2\alpha_s C_F}{\pi(\alpha - \beta)} \frac{(t_\beta - t_\alpha)(\beta t_\alpha - \alpha t_\beta)}{\beta} \right) e^{-\alpha_s \frac{C_F}{\pi} \left(\frac{t_\beta^2}{\beta} + \frac{(t_\alpha - t_\beta)^2}{\alpha - \beta} \right)} \Theta_{\alpha\beta}$$

**This log is
unaccounted for**

These logs
match

But the LL *does* live within the RDF's
undetermined part!
If we choose this higher-order wiggle

$$f_\alpha^{\text{LL}} = \frac{\beta}{\alpha t_\beta - \beta t_\alpha} \Theta_{\alpha\beta} + \frac{2\alpha_s C_F}{\pi(\alpha - \beta)} (t_\alpha - t_\beta) \Theta_{\alpha\beta}$$

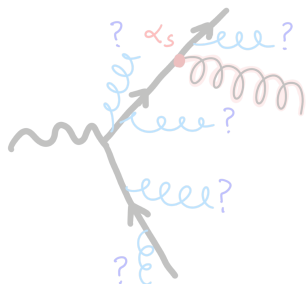
We would reproduce the LL.

So, it lives within the “theory
uncertainty” of the RDF.

We'll make theory uncertainties more
precise in a second!

The RDF is *not* guaranteed to be a full logarithmic resummation! **This log** is only first visible at $\mathcal{O}(\alpha^2)$, so the RDF matched to $\mathcal{O}(\alpha^1)$ missed it.

Outline



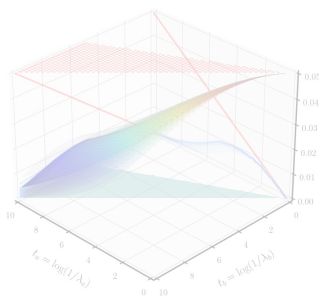
0. Basic Premise

Goal: Parameterize higher-order terms for the distribution of an observable

$$q(t|\alpha) = f(t, \alpha) \cdot \exp \left[- \int_0^t dt' f(t', \alpha) \right]$$

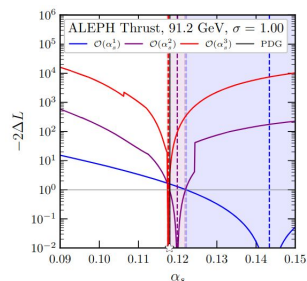
1. The Resummed Distribution Function

Given a fixed order perturbative calculation, write down a form for all possible higher order corrections consistent with unitarity



2. QCD, Examples, and Analyses

Connections to Resummation, QCD observables, numeric matching, and discussions.



3. Theory Uncertainties, Nuisance Parameters, and Fits

Fits to ALEPH data, α_s extraction, theory nuisance parameters

α_s Extraction from Data

Once we have our prediction for our observables t given α , let's compare it to data!

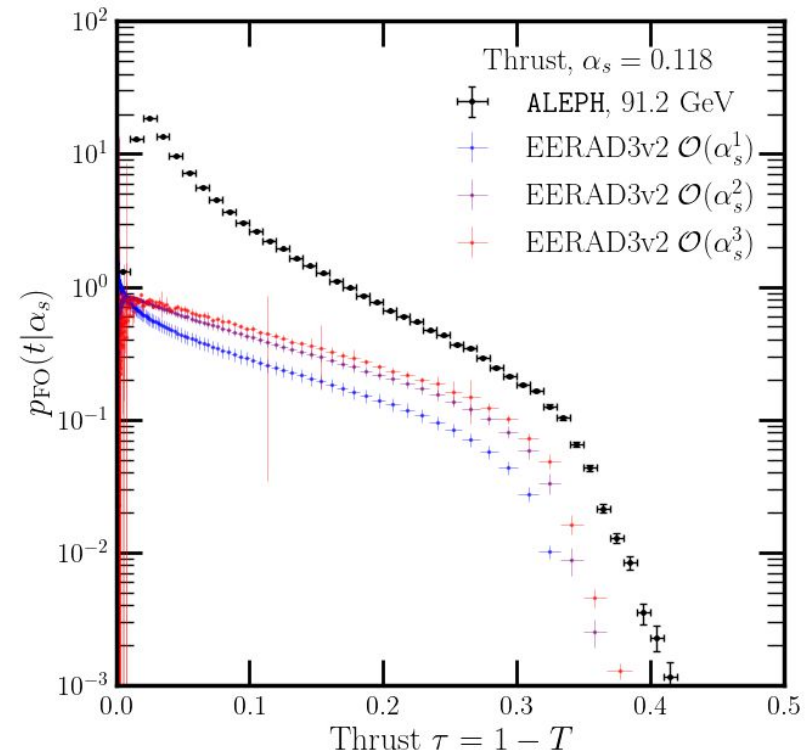
ALEPH (LEP) measurements for thrust in e^+e^- collisions at 91.2 GeV

Goal: given data t , find the α that maximizes $q(t | \alpha)$ — **Maximum Likelihood Estimation (MLE)**

The data is binned, so we will use a binned MLE. If we had access to event-level data, we could do an unbinned MLE, since the RDF is directly a valid likelihood!

Ill-defined MLE for fixed order calculations without windows or regulators.

ALEPH vs. Fixed Order Predictions



*I am a theorist and therefore I am going to completely ignore systematic uncertainties.

Theory Uncertainties and Nuisance Parameters

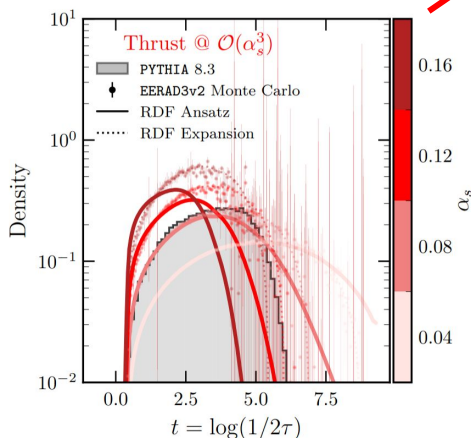
Theory uncertainties are about trying to **quantify what we don't know**.
But, we've already parameterized the physics we don't know with g coefficients!

Promote to **Nuisance Parameters**

The same split
we've been doing
this whole time

$$g(t, \alpha) = \underbrace{g_{\text{Matched}}(t, \alpha, \phi)}_{\mathcal{O}(\alpha_s^M)} + \underbrace{g_{\text{Nuisance}}(t, \alpha; \nu)}_{\mathcal{O}(\alpha_s^{M+1})}$$

Derived
from



Then, just fit the profile likelihood to data:

$$\log \mathcal{L}(\alpha) = \log \frac{\mathcal{L}(\alpha, \hat{\nu})}{\mathcal{L}(\hat{\alpha}, \hat{\nu})}$$

Emotionally: Statistical power gets “eaten up” fitting **nuisances**, leading to a “theory uncertainty” on α

Theory Uncertainties and Nuisance Parameters

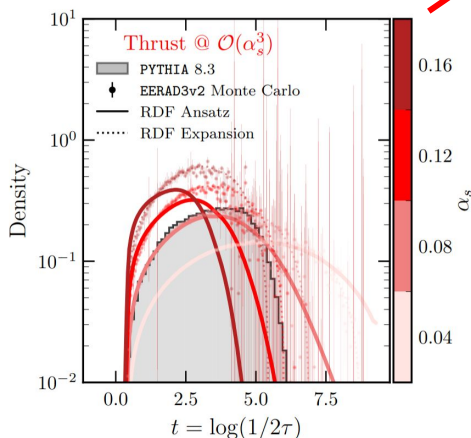
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Derived
from



Very similar in flavor to **Theory Nuisance Parameters** (TNPs), where theory uncertainties are also derived by profiling over higher order corrections.

We don't assume any^{*} physics structure however, but the RDF *does* enforce unitarity.

^{*}Caveat: We have to assume *something* to avoid infinite degeneracy. Next slide!

[Aside] Theory Uncertainties and Priors

Philosophically, all theory uncertainties attempt to model our ignorance about higher-order terms.

Scale Variations attempt to do this by varying renormalization scales, which affects higher order terms. But it assumes that higher orders look like lower orders.

TNPs make some broad (and probably reasonable) assumptions about the form of these higher order terms

Other approaches [Cacciari, Houdeau; [1105.5152](#)] use boundedness / truncation properties.

RDFs make no assumptions^{*}, but if the higher order terms can truly be anything, all statistical power gets absorbed away — *not even wrong!*

We need a regulator^{**} or a prior on coefficients to break degeneracies.

$$\frac{1}{2} \sum_k \frac{|\nu_k - \mu_k|^2}{\sigma_k^2}$$
$$\sigma_k = \frac{\sigma}{m!n!} \quad \mu_k = 0$$

Let's just put a reasonable Gaussian prior on g and not think about it too hard or fine-tune^{***} it.

Pontificating: Theory uncertainties are inherently Bayesian as higher-order terms aren't probabilistic. That's why I said "semi-frequentist" earlier.

^{*}Other than unitarity

^{**}The fact that the RDF only has finitely many parameters is a kind of regularization, but it is extremely weak.

^{***}You can always fine-tune a prior to give almost any answer you want. We consider it a feature of the RDF that simple unit Gaussians work well.

[Details] Fitting

Try our repo!



We use a χ^2 binned likelihood:

$$-\log \mathcal{L}(\alpha, \nu) = \frac{1}{2} \sum_{\text{Bin}_i} \frac{|\text{RDF}(\text{Bin}_i, \alpha, \phi, \nu) - \text{data}(\text{Bin}_i)|^2}{\text{error}(\text{Bin}_i)^2} + \frac{1}{2} \sum_k \frac{|\nu_k - \mu_k|^2}{\sigma_k^2}$$

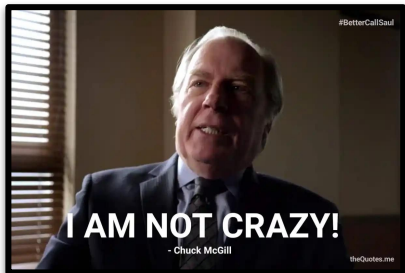
For each α in a fine-grained scan, we fit for the optimal **order-(M+1) g nuisance parameters** that minimize the likelihood, keeping the matched **order-(M) and below g parameters** frozen.

The best fit α is then the one that minimizes the profiled log likelihood.

We can construct confidence intervals in the region:

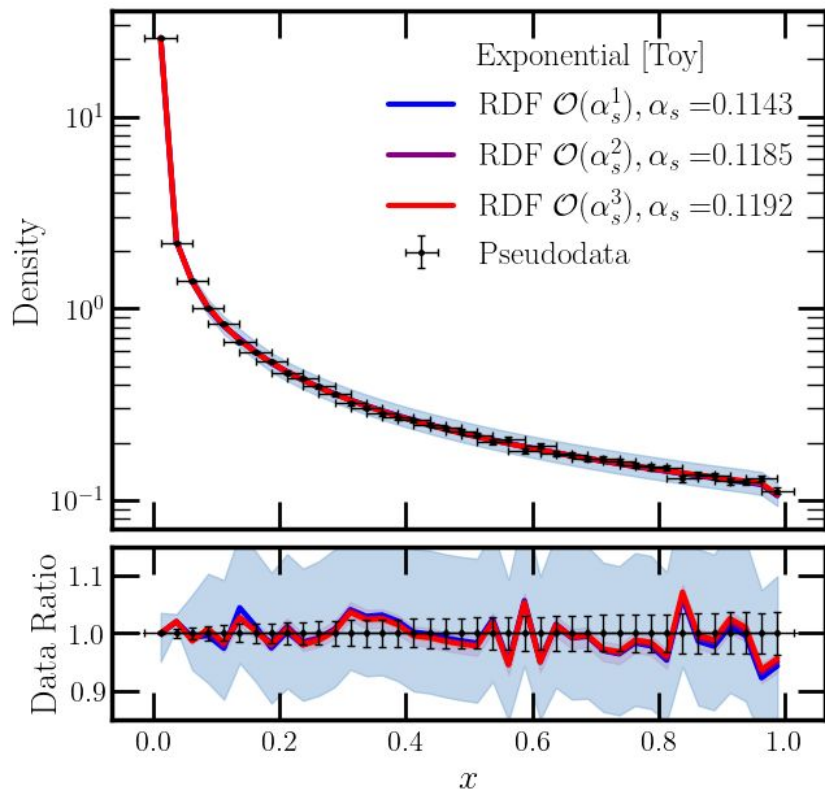
$$-2 \log \mathcal{L}(\alpha) < 1$$

We don't check that our intervals are calibrated / check for Wilke's Theorem.
But in a real analysis, you should!

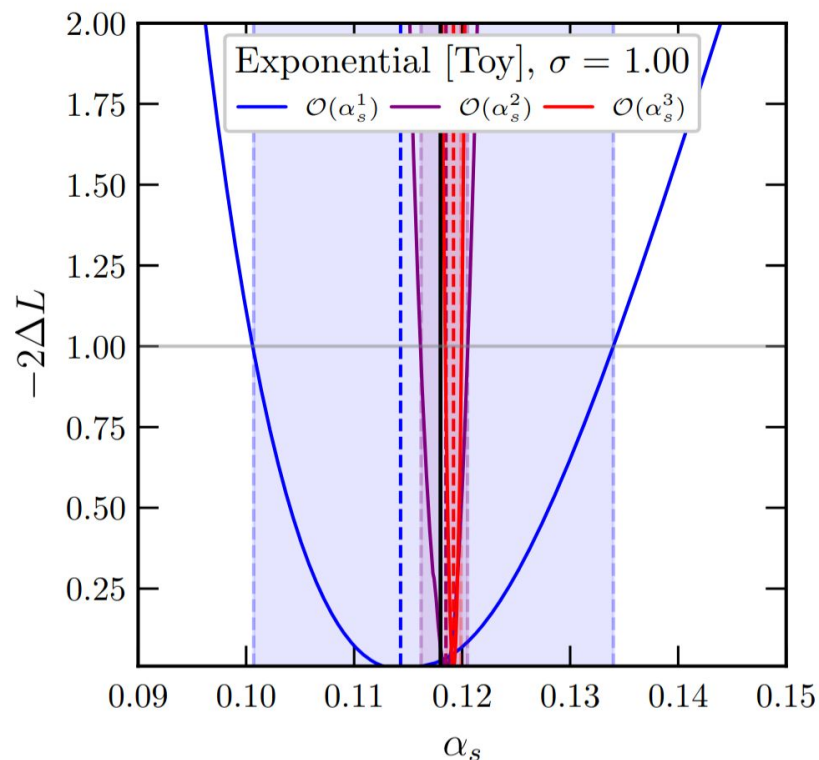


Proof of Concept: Exponential

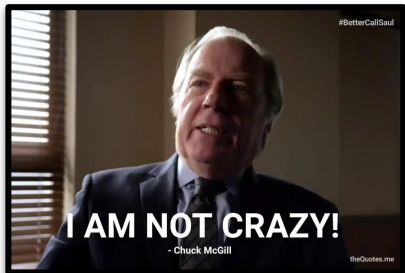
As a sanity check, let's check that we get reasonable results on a toy before real QCD.



Applying our RDF fitting procedure to toy data (**match** to F.O., then **fit** to data)

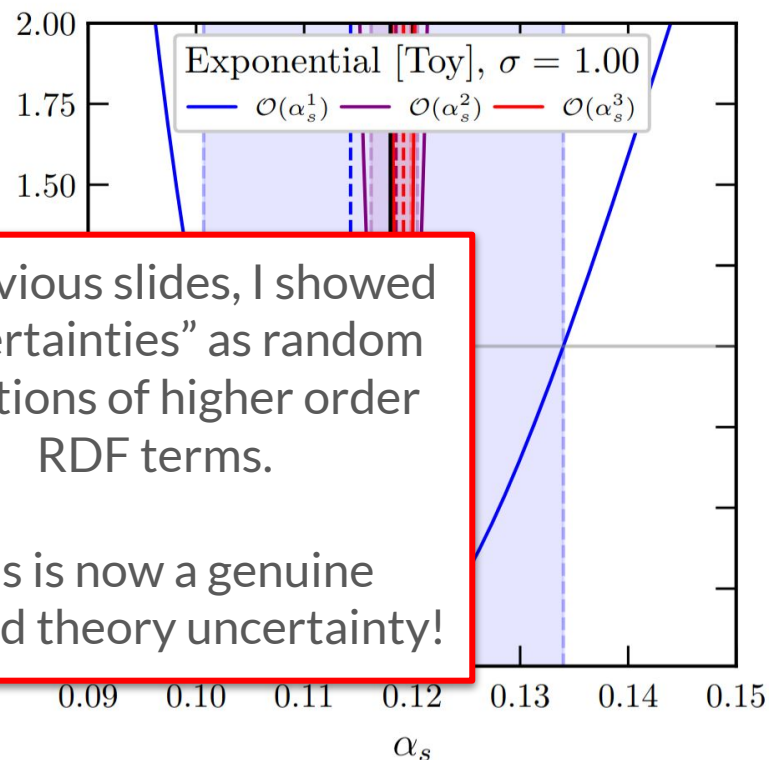
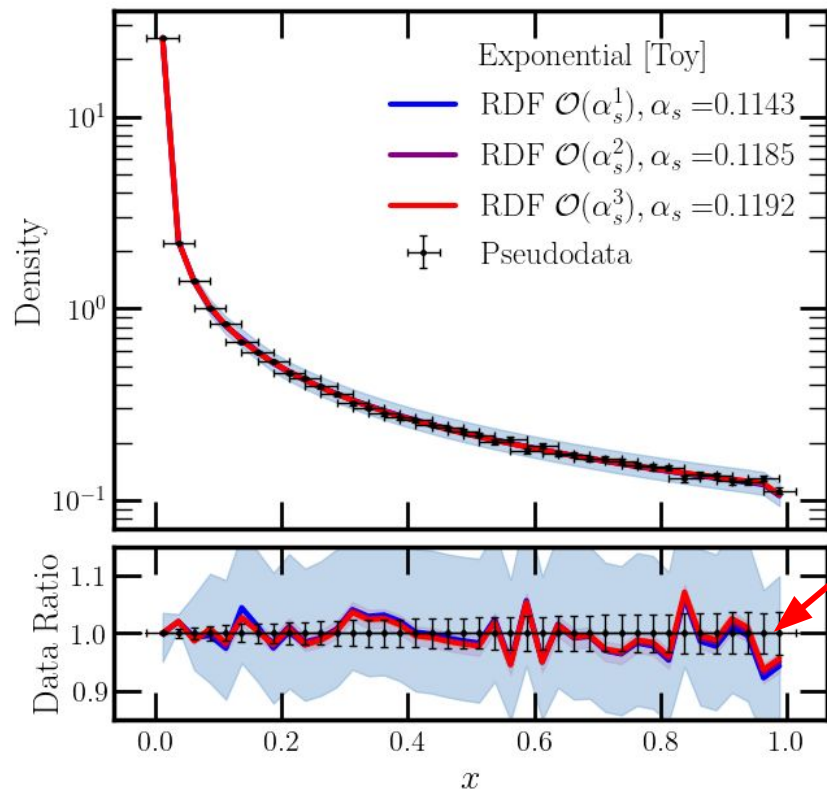


We can extract genuine and correct-looking confidence intervals for α !



Proof of Concept: Exponential

As a sanity check, let's check that we get reasonable results on a toy before real QCD.



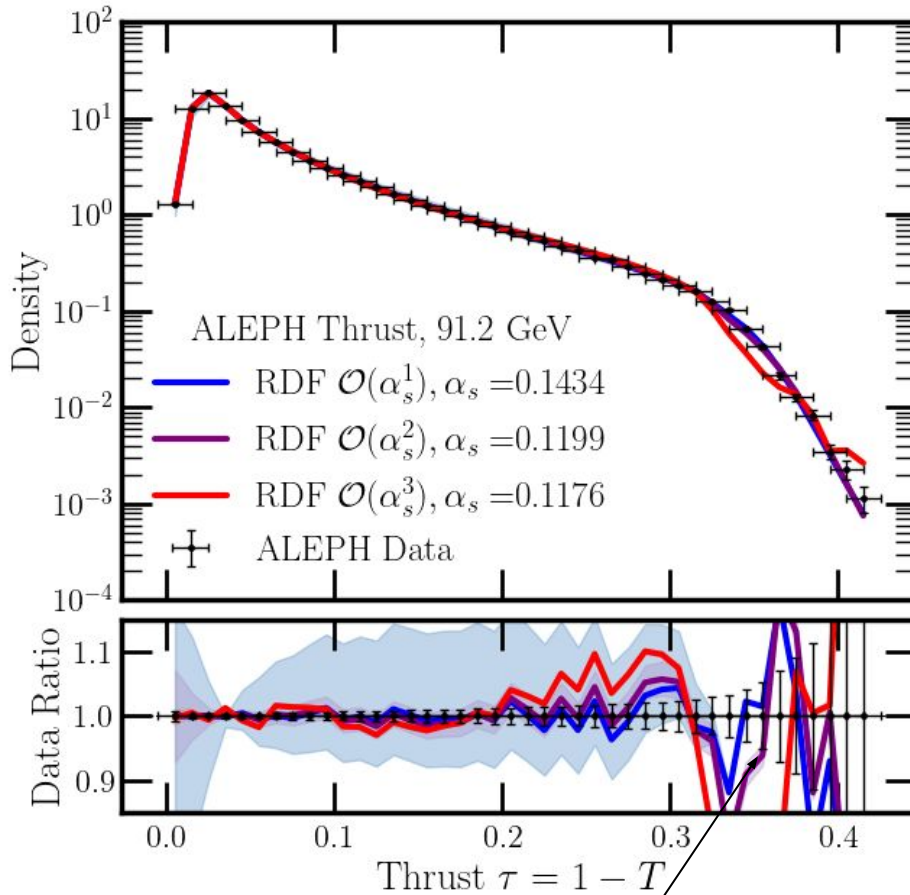
In previous slides, I showed “uncertainties” as random variations of higher order RDF terms.

This is now a genuine profiled theory uncertainty!

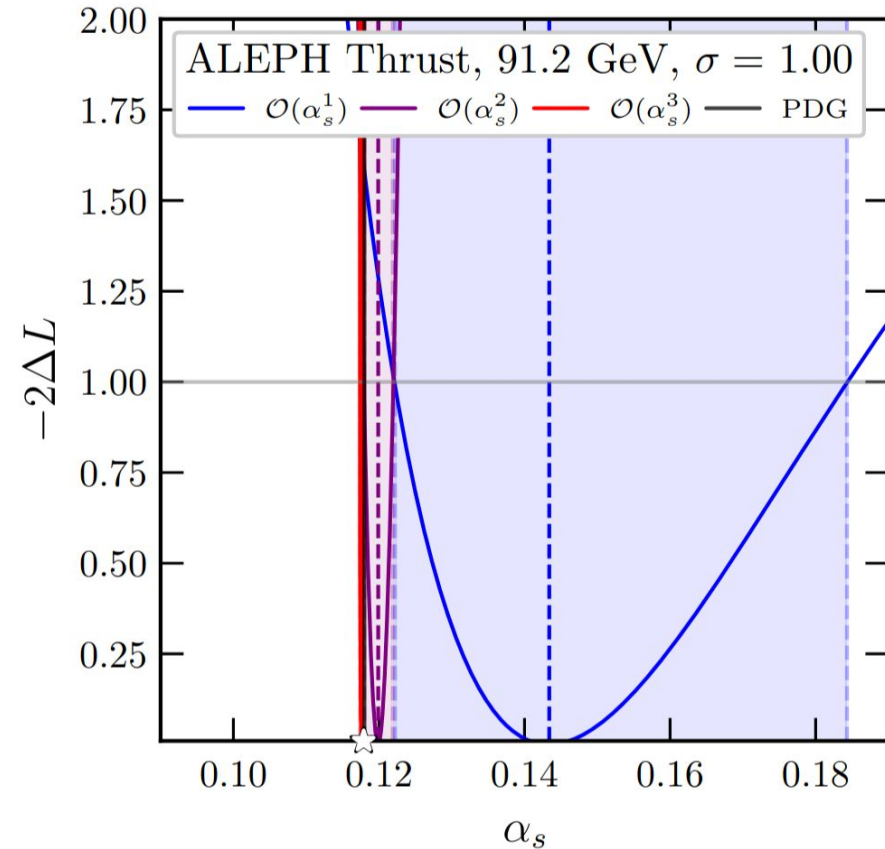
Applying our RDF fitting procedure to toy data (match to F.O., then fit to data)

We can extract genuine and correct-looking confidence intervals for α !

Fitting Thrust



Kinematic thresholds are tricky to fit...



Beyond leading order, we recover the PDG value. **Victory!**

Final Results

Just to be thorough, let's try widening and narrowing our prior to see if it drastically changes things. *Pretty robust, at least beyond leading order!*

Order	<i>RDF Prior σ</i>			LEP I+II (From Ref. [40])
	$\sigma = 0.5$	$\sigma = 1.0$	$\sigma = 2.0$	
Thrust $\mathcal{O}(\alpha_s^1)$	$0.1555^{+0.0189}_{-0.0139}$	$0.1434^{+0.0409}_{-0.0212}$	$* 0.19^{+0.0000}_{-0.0595}$	0.1142 ± 0.0297
Thrust $\mathcal{O}(\alpha_s^2)$	$0.1202^{+0.0010}_{-0.0009}$	$0.1199^{+0.0019}_{-0.0020}$	$0.1249^{+0.0067}_{-0.0053}$	0.1152 ± 0.0068
Thrust $\mathcal{O}(\alpha_s^3)$	$0.1202^{+0.0002}_{-0.0001}$	$0.1176^{+0.0002}_{-0.0002}$	$0.1164^{+0.0003}_{-0.0002}$	0.1164 ± 0.0033

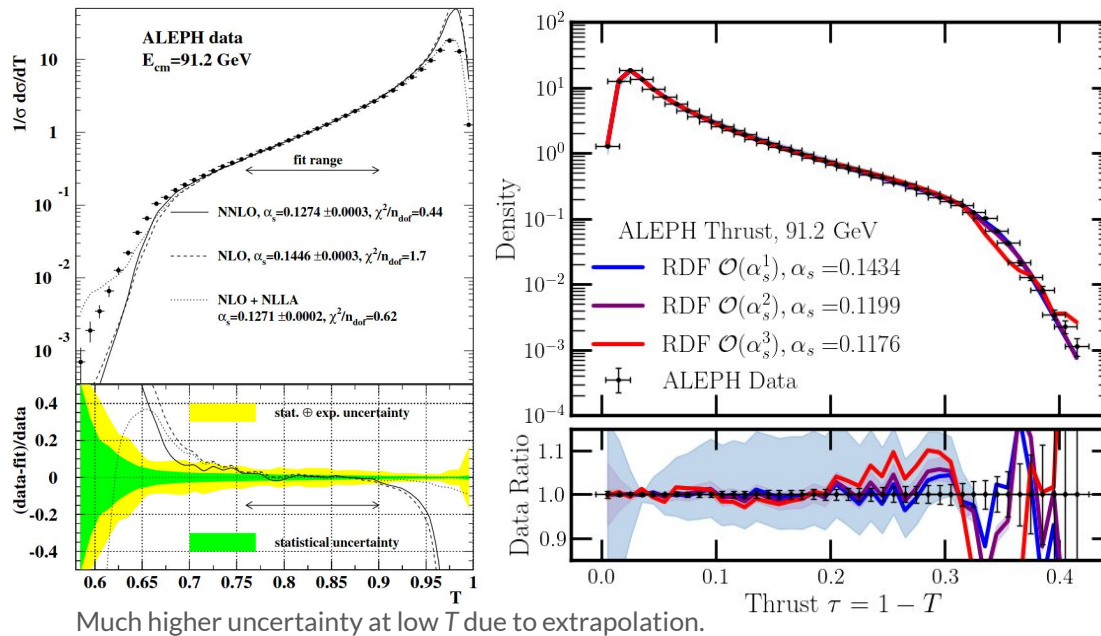
For comparison, a proper SCET calculation of thrust fit to the same data (but windowed) with scale variation uncertainties. Same ballpark!

At leading order, it seems there isn't enough information for us to reliably extract α .

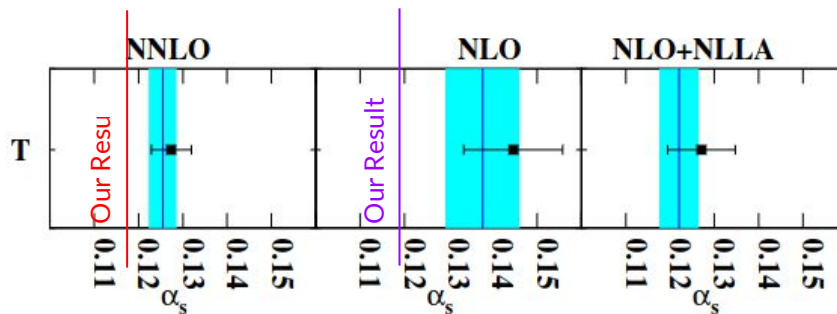
*The third order errorbars are smaller than the SCET calculation. This is likely due to how we chose our prior to scale. A little fine tuning might help a lot!
Remember we have *not* considered systematic uncertainties or nonperturbative uncertainties here.

For Comparison – Just Fixed Order

[Dissertori, Gehrmann–De Ridder, Gehrmann, Glover, Heinrich, Stenzel; [0712.0327](#)]



Much higher uncertainty at low T due to extrapolation.



To fit just a pure fixed order calculation, use only a window.

Pure fixed order results are very different! α goes to 0.125 or even higher.

The **RDF**, *essentially for free*, obtains results consistent with the PDG average, *still only using fixed order info!*

All we did was re-organize the fixed order info in a motivated way.

[[0712.0327](#). Figure slightly modified to remove other observables]

CLAIMER

What we are claiming

- **Claiming:** Given a distribution p , pathological or otherwise, we can find a regulated distribution q that is normalized, smooth, positive, and normalized that is perturbatively matched to p .
- **Claiming:** This is a universal approximator: *any* valid q can be reached with a choice of higher order g , including the “true” one or N^nLL calculations.
- **Claiming:** For *particular* g ’s, we can get meaningful q ’s that look like theory calculations and tame large logs!
- **Claiming:** Higher order g ’s can be used as meaningful nuisance parameters to define theory uncertainties and do fits!

DISCLAIMER

What we are *not* claiming

- **Not Claiming:** The perturbative structure is *easy* to maintain numerically or that numeric artifacts are manageable.
 - But if you restrict to reasonable choice or priors of g , like $g \sim N(0,1)$, then the results seem reasonable!
- **Not Claiming:** The envelopes give a *prior-free* uncertainty on higher orders.
 - The envelopes only tell you what *possible* higher order physics is consistent with positivity and normalizability — it *constrains* physics!
- **Not Claiming:** This is or beats N^nLL resummation.
 - But it is an all-orders resummation, and in special cases really is N^nLL !
 - For the RDF you don’t need to know *any* physics! Potentially useful for joint distributions
 - Even if you have a logarithmic resummation already, the RDF can still help you do theory uncertainties!

Key Point: This is *not* a shortcut to doing QCD, just a new way to parameterize things to guarantee nice properties. There is no free lunch, you still have to do physics!

Future Directions and Ideas

- (1) We chose the simplest possible polynomial parameterization and the simplest priors

$$g^*(t, \alpha), \quad g_{\text{Analytic}}(t, \alpha) = \sum_{m=0, n=0}^{M, N} g_{mn} \alpha^m t^n \Theta(t - \theta_{mn})$$

$$\frac{1}{2} \sum_k \frac{|\nu_k - \mu_k|^2}{\sigma_k^2}$$

Surely its possible to do something smarter with a little thought!

- (2) We explicitly ignored resurgence / asymptotic series / nonperturbative effects.

$$g_{\text{RDF}} \stackrel{?}{=} \sum_m \alpha^m g_m + \cancel{\frac{1}{\alpha} g_{\text{asympt.}}} + \cancel{g_{\text{non. pert.}}}$$

Surely there is a clever way to parameterize these nicely!

- (3) We didn't talk about enforcing RG equations or SCET factorization with or within the RDF. Surely you can inject some QCD knowledge this way!

Future Directions and Ideas [Cont'd]

We took the **physics we know** to be a fixed order calculation, and the **physics we don't know** to be higher orders.

Diagram illustrating the matrix g_{mn} structure, showing the relationship between Powers of logs t (horizontal axis) and Orders of α (vertical axis).

	g_{00}	g_{01}	g_{02}	g_{03}	g_{04}	g_{05}	g_{06}	g_{07}	g_{08}	g_{09}
	g_{10}	g_{11}	g_{12}	g_{13}	g_{14}	g_{15}	g_{16}	g_{17}	g_{18}	g_{19}
	g_{20}	g_{21}	g_{22}	g_{23}	g_{24}	g_{25}	g_{26}	g_{27}	g_{28}	g_{29}
	g_{30}	g_{31}	g_{32}	g_{33}	g_{34}	g_{35}	g_{36}	g_{37}	g_{38}	g_{39}
	g_{40}	g_{41}	g_{42}	g_{43}	g_{44}	g_{45}	g_{46}	g_{47}	g_{48}	g_{49}

The matrix is divided into two regions:

- Known:** The upper-left region (pink shading), corresponding to lower orders of α and lower powers of logs t .
- Unknown:** The lower-right region (blue shading), corresponding to higher orders of α and higher powers of logs t .

Surely there are other ways to split what we know and what we don't! Perhaps we know some logs (fitting Pythia or Panscales) and don't know others?

Diagram illustrating the matrix g_{mn} structure, showing the relationship between Powers of logs t (horizontal axis) and Orders of α (vertical axis).

	g_{00}	g_{01}	g_{02}	g_{03}	g_{04}	g_{05}	g_{06}	g_{07}	g_{08}	g_{09}
	g_{10}	g_{11}	g_{12}	g_{13}	g_{14}	g_{15}	g_{16}	g_{17}	g_{18}	g_{19}
	g_{20}	g_{21}	g_{22}	g_{23}	g_{24}	g_{25}	g_{26}	g_{27}	g_{28}	g_{29}
	g_{30}	g_{31}	g_{32}	g_{33}	g_{34}	g_{35}	g_{36}	g_{37}	g_{38}	g_{39}
	g_{40}	g_{41}	g_{42}	g_{43}	g_{44}	g_{45}	g_{46}	g_{47}	g_{48}	g_{49}

The matrix is divided into two regions:

- Known:** The lower-left region (pink shading), corresponding to lower orders of α and lower powers of logs t .
- Unknown:** The upper-right region (blue shading), corresponding to higher orders of α and higher powers of logs t .

Future Directions and Ideas [Cont'd Cont'd]

Questions I have for you!

- Are there non-QCD (general QFT, or astrophysical/cosmological) applications where distributions are pathological due to incomplete information and the RDF can be applied to tame them?
- In EFTs and the S-matrix program, unitarity is used to bound higher order Wilson coefficients and amplitudes. Could the RDF mesh with these types of bounds, working at the amplitude level rather than cross section level?
- We have only use unitary in fact, not even all of unitarity as a physical principle. Are there other interesting physical principles it would be useful for the RDF to constrain terms with?

Conclusion

Try our repo!



Email us questions!
gambhirb@ucmail.uc.edu
rmastand@uchicago.edu
[RG, Mastandrea; 2512.04160]



We introduced the Resummed Distribution Function (RDF) as a method to parameterize non-pathological distributions, that splits **physics we know** from **physics we don't** (high order)

$$p(x|\alpha) = p_0(x) + \alpha p_1(x|\alpha) + \alpha^2 p_2(x|\alpha) + \dots + \alpha^M p_M(x|\alpha) + \mathcal{O}(\alpha^{M+1})$$

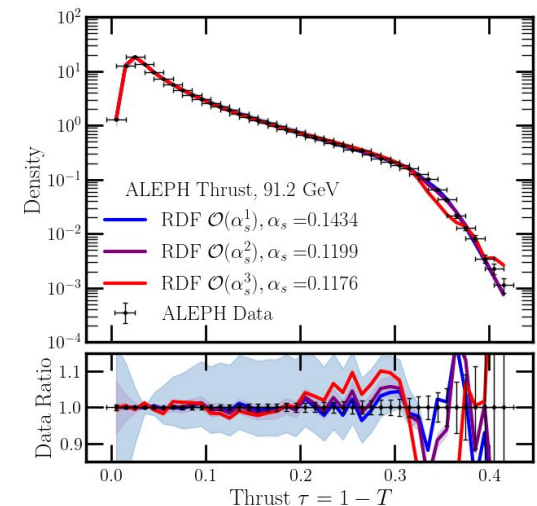
What I know: Fixed-order calculation

Unknown

It is a completely agnostic, novel, and easy way to convert a fixed-order distribution to a unitary resummation, containing all possible all-orders completions

The RDF provides a natural and rigorous way to place theory uncertainties on perturbative calculations!

If you have a fixed-order calculation, zero reason not to apply the RDF to explore higher orders for free!



Appendices

Numerics and Fitting

Bogosort

Article Talk

From Wikipedia, the free encyclopedia

In computer science, **bogosort**^{[1][2]} (also known as **permutation sort** and **stupid sort**^[3]) is a sorting algorithm based on the generate and test paradigm. The

We use a “**0-1-2**” strategy for numeric minimization, based on how many derivatives are considered:

0th Order: “BOGO-init” We randomly initialize all parameters N times. We select the lowest loss. Then we randomly perturb each of those parameters again with lower variance, and repeat many times. *Not too dissimilar from Hamiltonian Monte Carlo*

1st Order: Gradient Descent We use ordinary gradient descent (specifically, **ADAM**) to minimize the parameters.

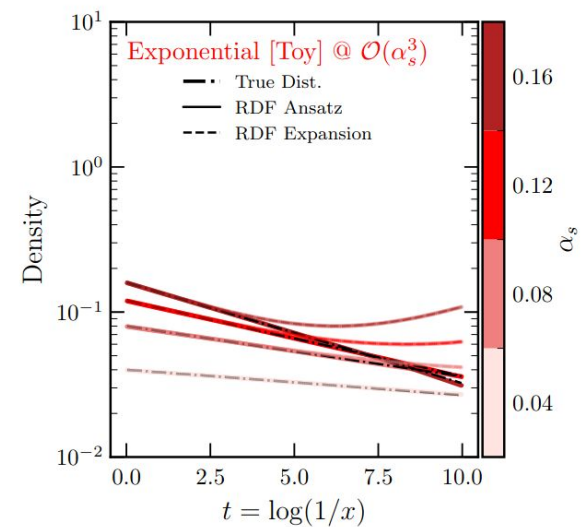
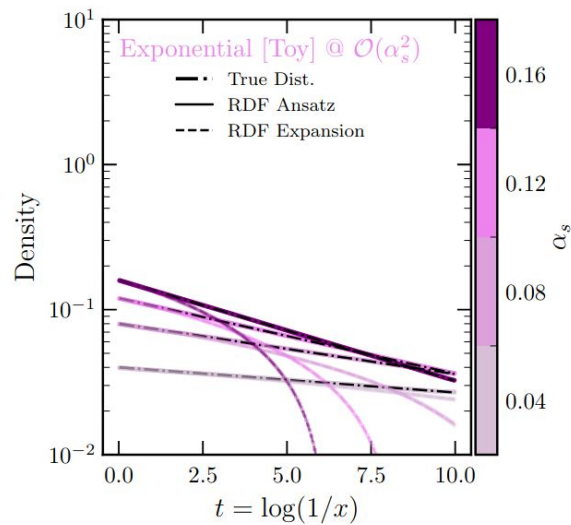
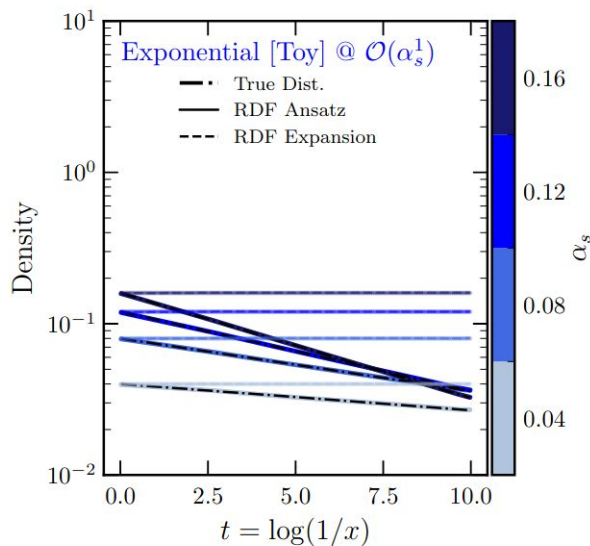
2nd Order: Refinement We use an approximate 2nd order minimizer (**LBFGS**) to refine our estimate.

“Machine Learning”?

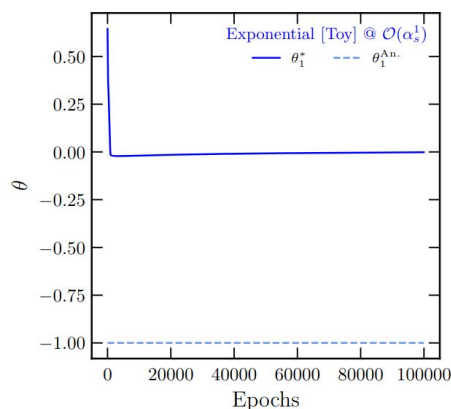
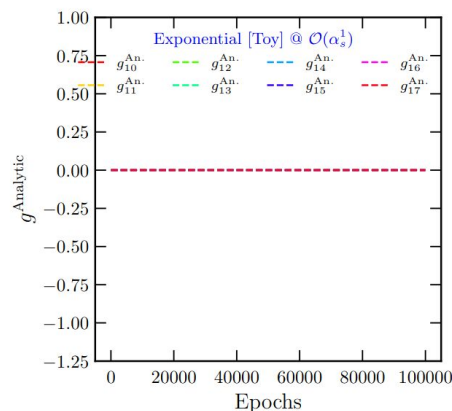
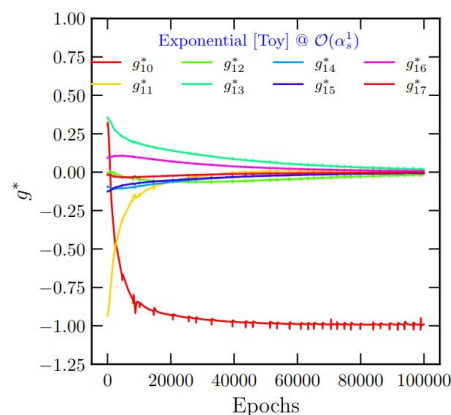
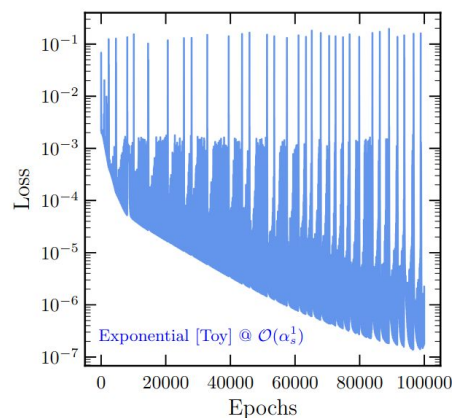
For each α in our scan, we seed the fit with the results of the previous α . We scan back and forth more than once to avoid hysteresis.

Numeric Matching for Toys

Dashed RDF Taylor Expansions = Fixed Order Exponential Expansions, so victory!



Numeric Matching for Toys



For the exponential, we know what the analytic matching coefficients should be, so we can verify the numerics:

$$g^*(t, \alpha) = \alpha$$

$$g^{\text{Analytic}}(t, \alpha) = 0 + \mathcal{O}(\alpha^M)$$